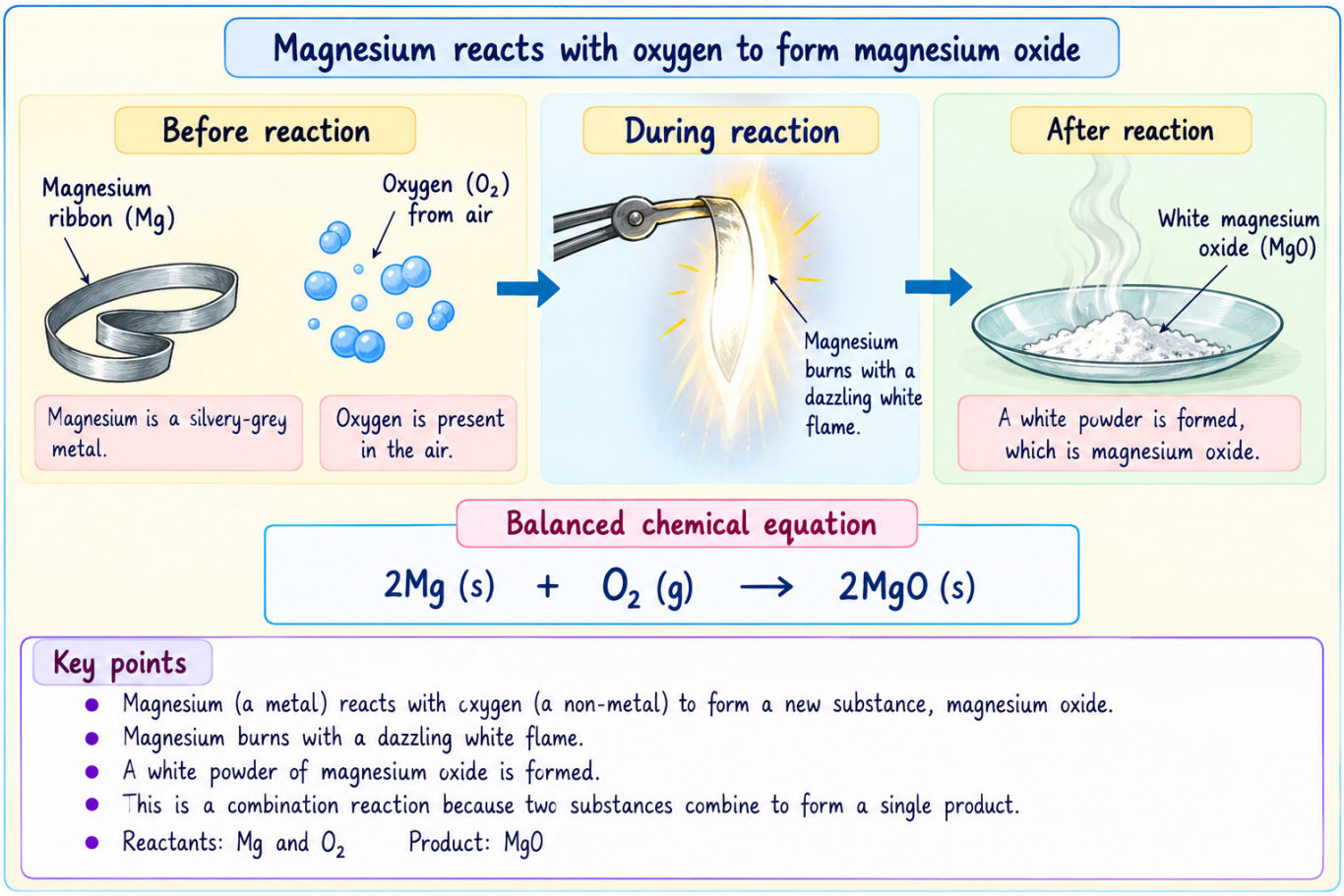
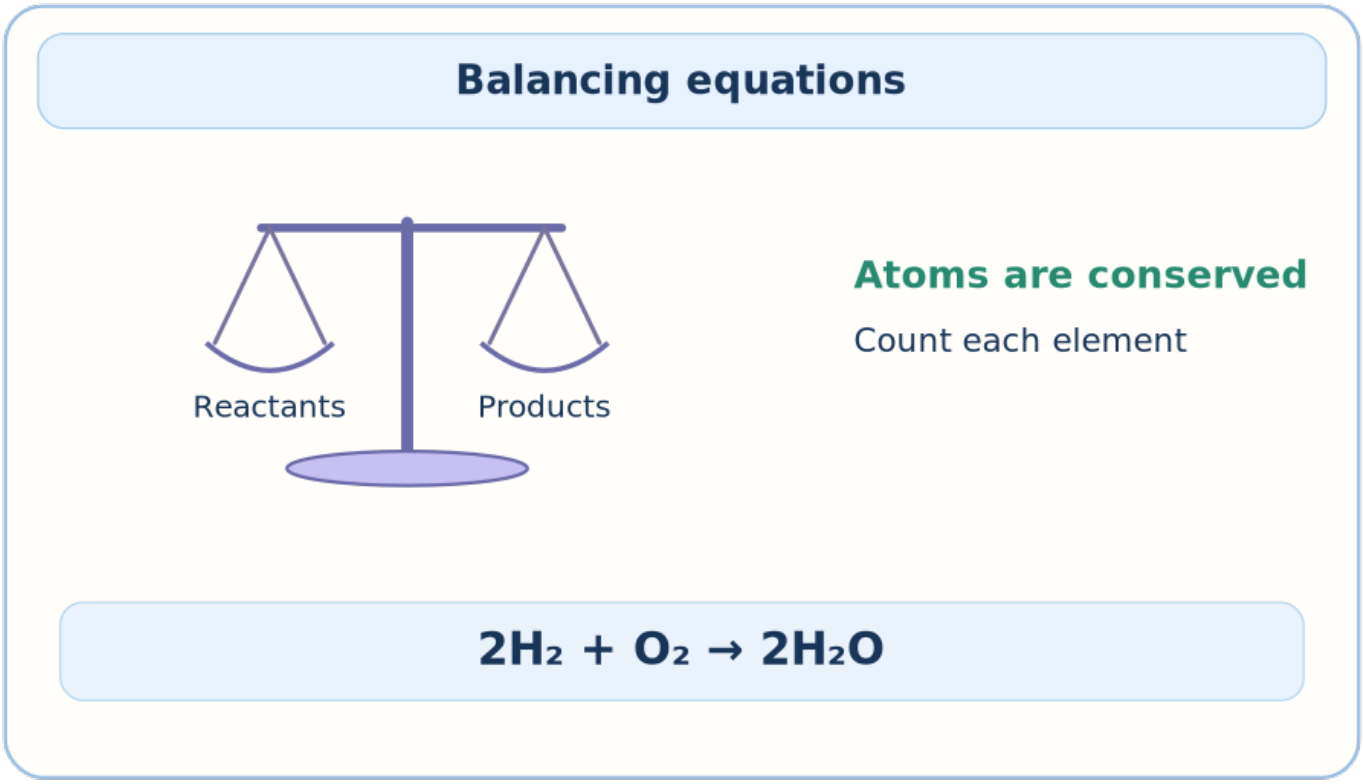
EduLLM — 62 Explanation Illustrations

Questions remain printed and interactive; these visuals appear after students answer. Four illustrations are detailed hand-drawn style PNGs, while 58 are simpler original vector drawings. Teacher review is recommended.

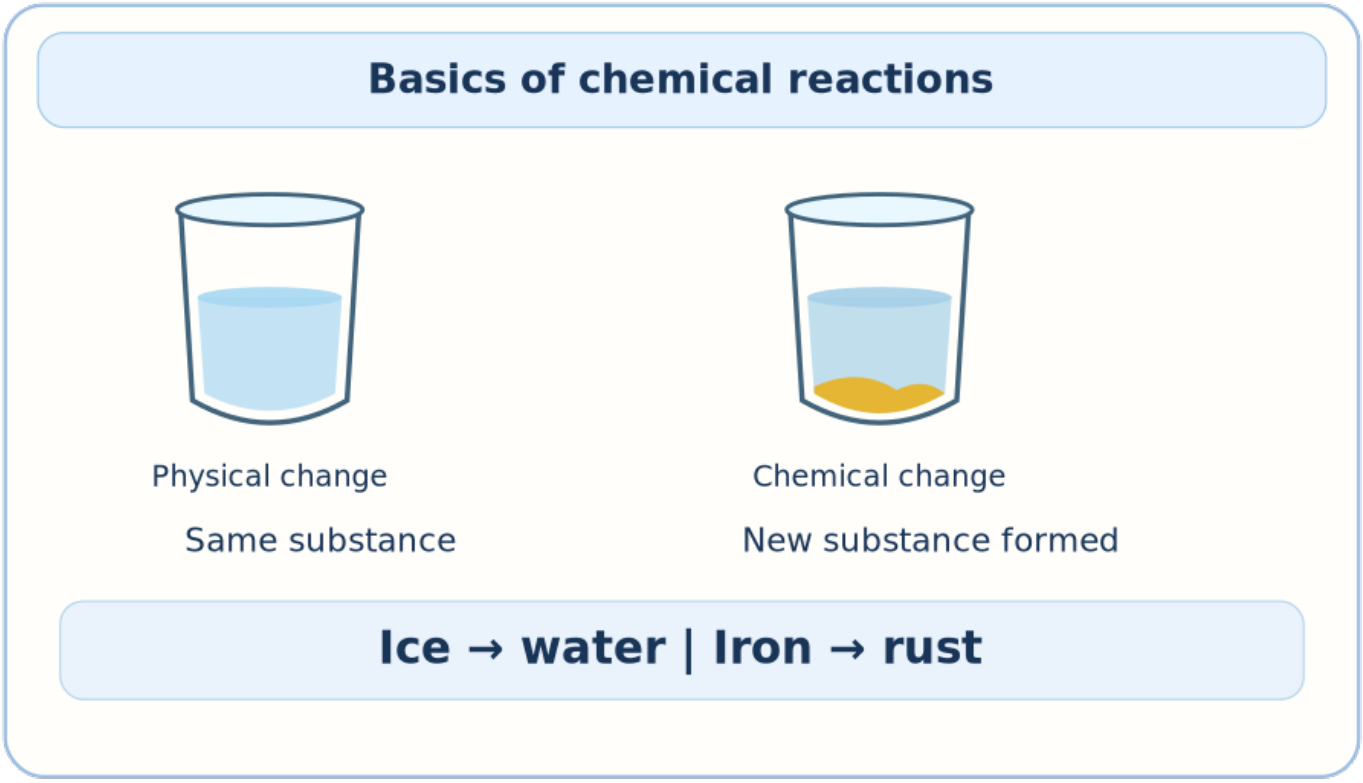
Q1. What is a chemical reaction?



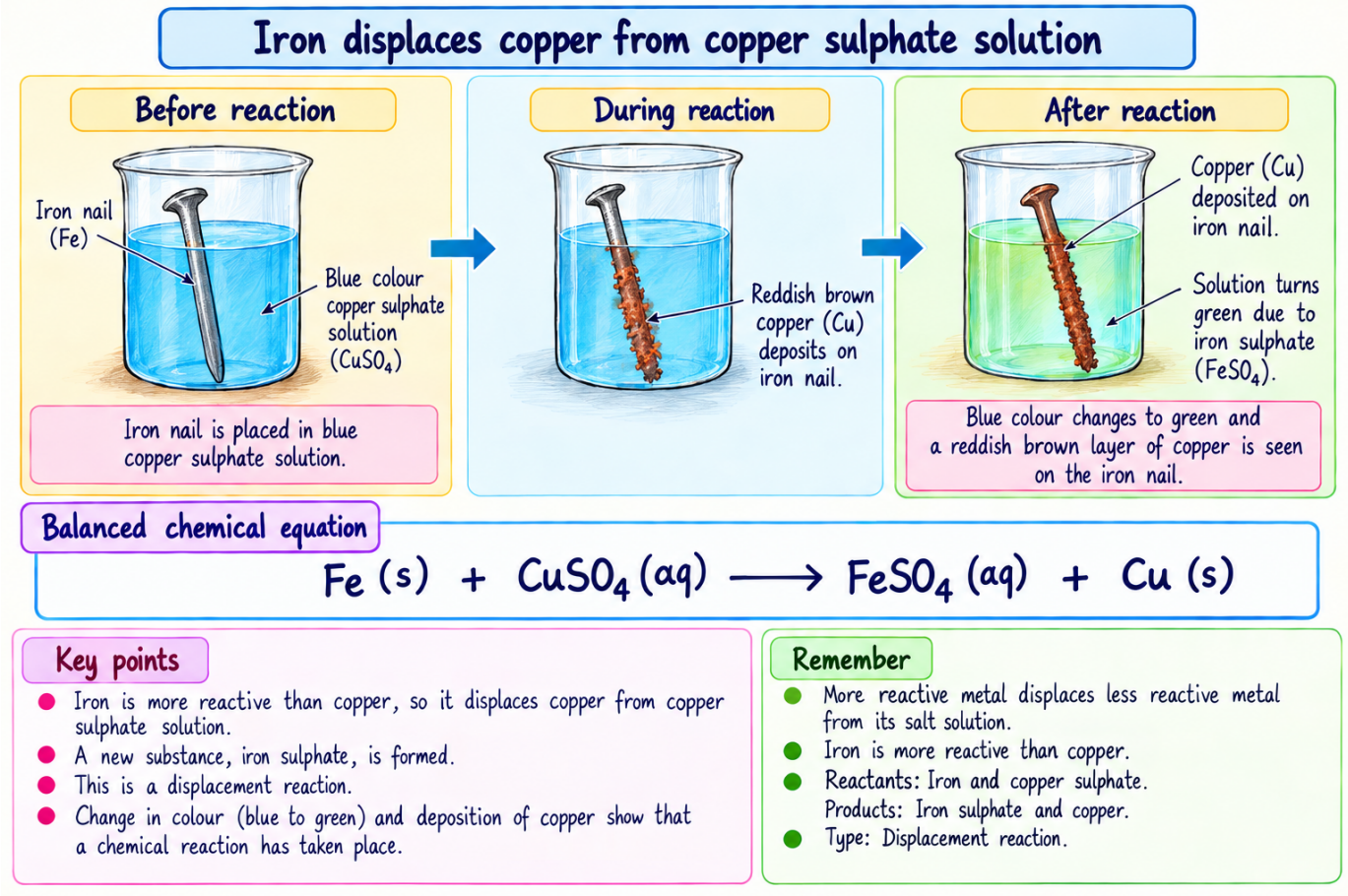
Q2. Why should a chemical equation be balanced?



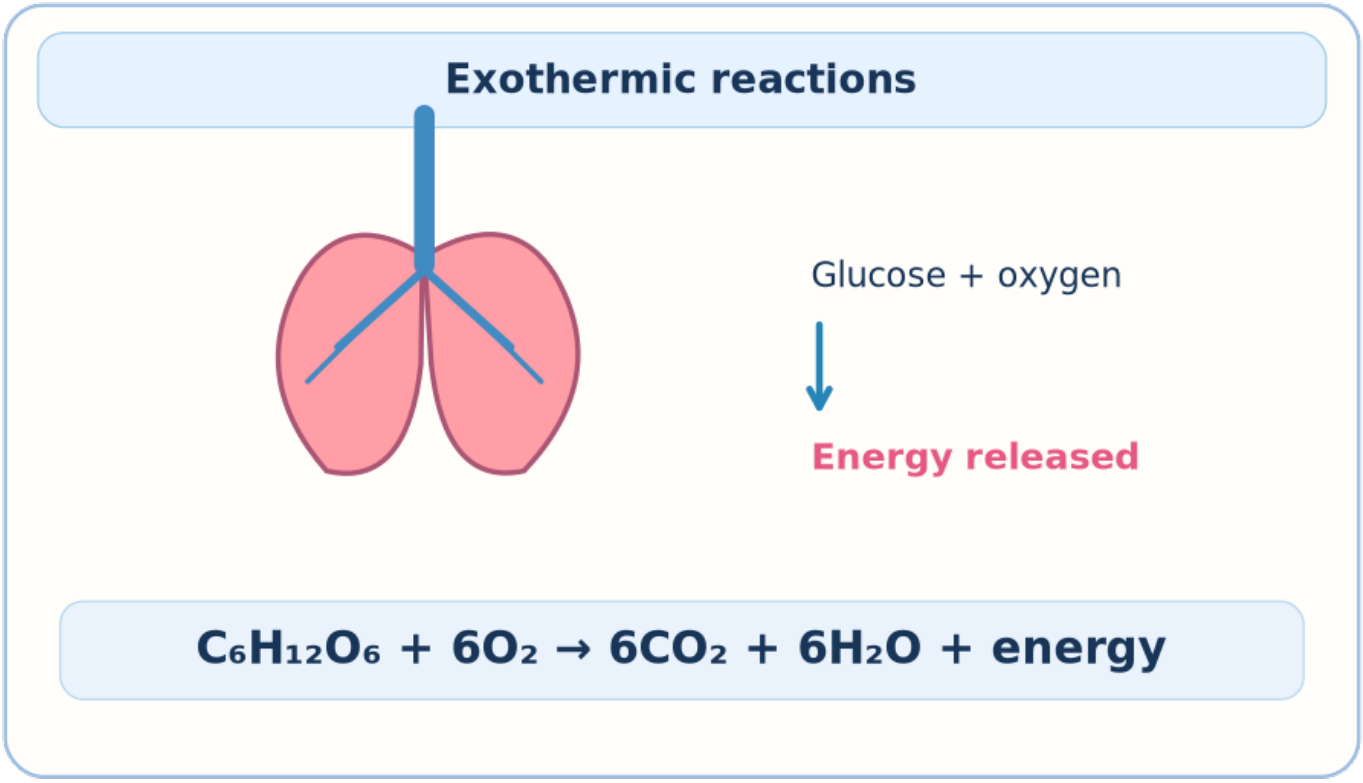
Q3. Differentiate physical change and chemical change, with examples.



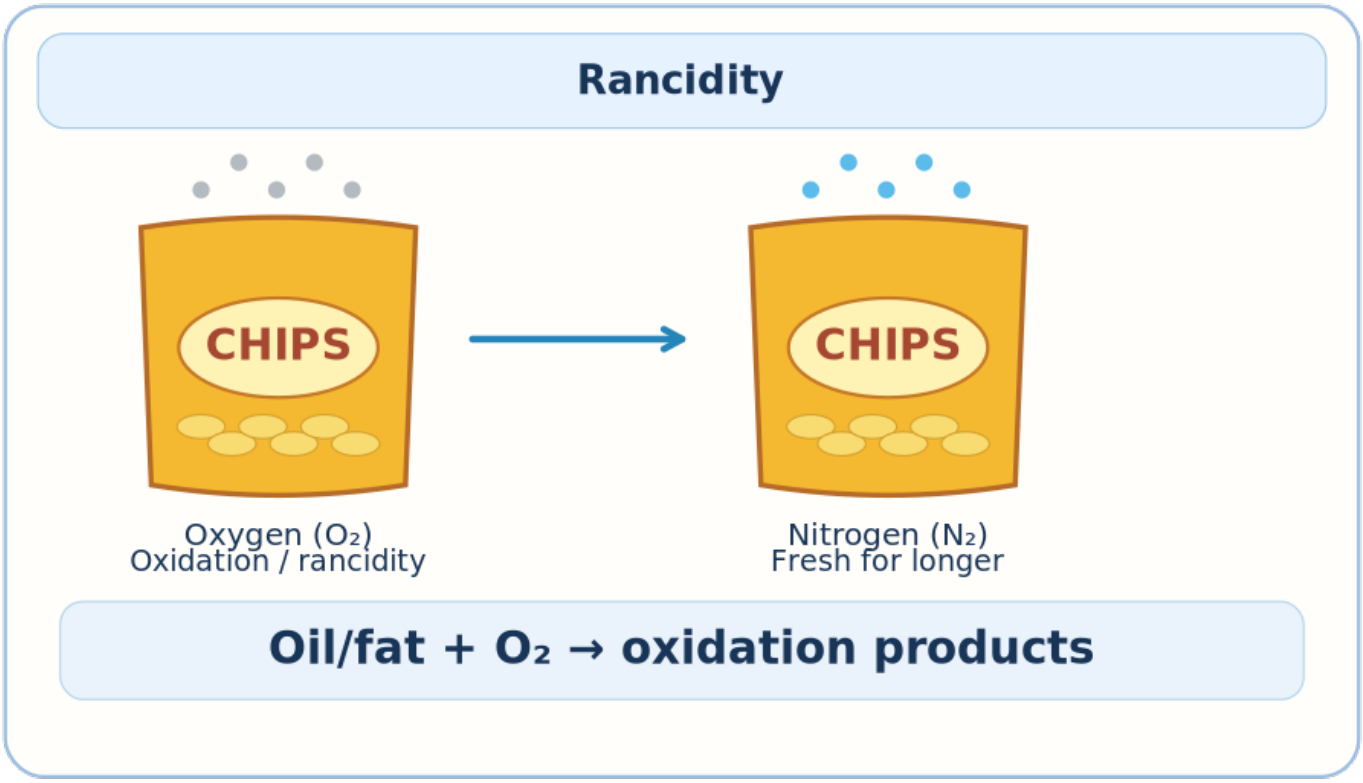
Q4. What happens when an iron nail is dipped into copper sulphate solution?



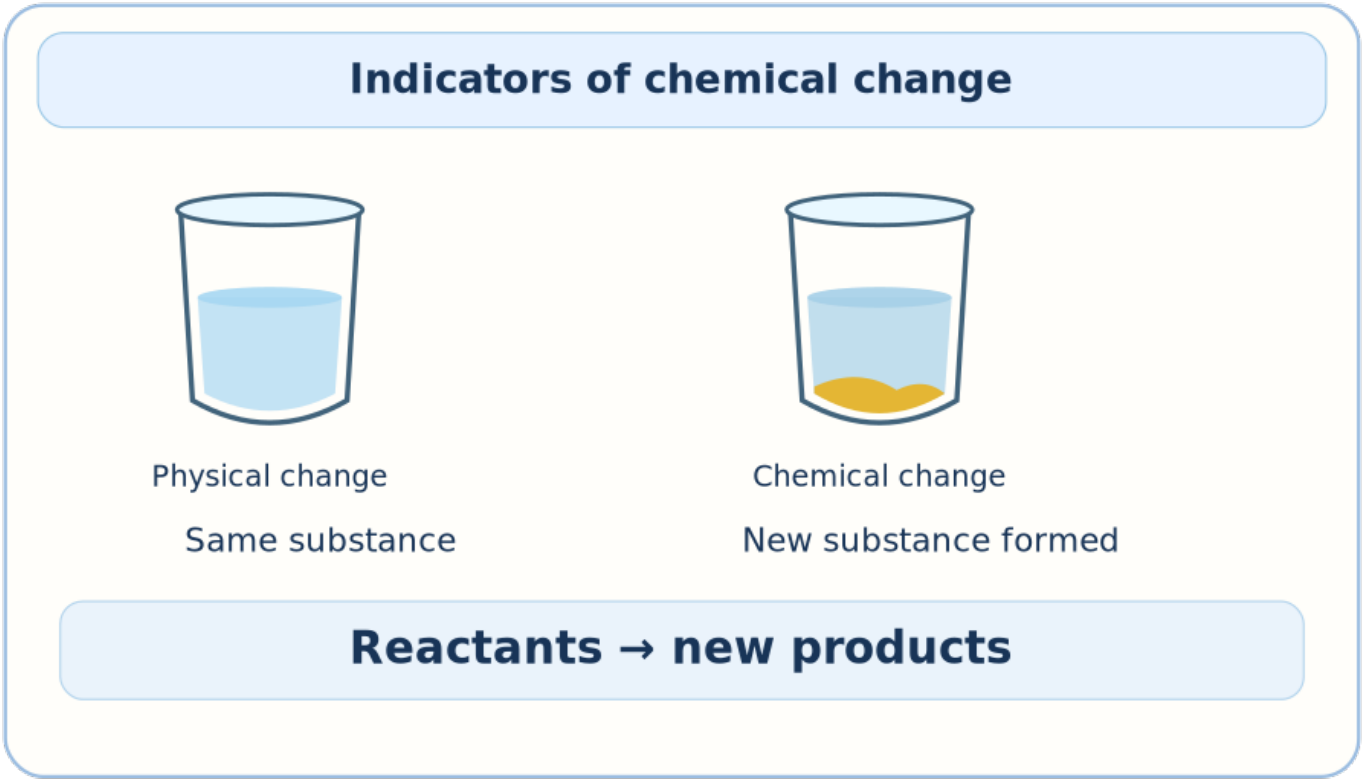
Q5. Why is respiration called an exothermic reaction?



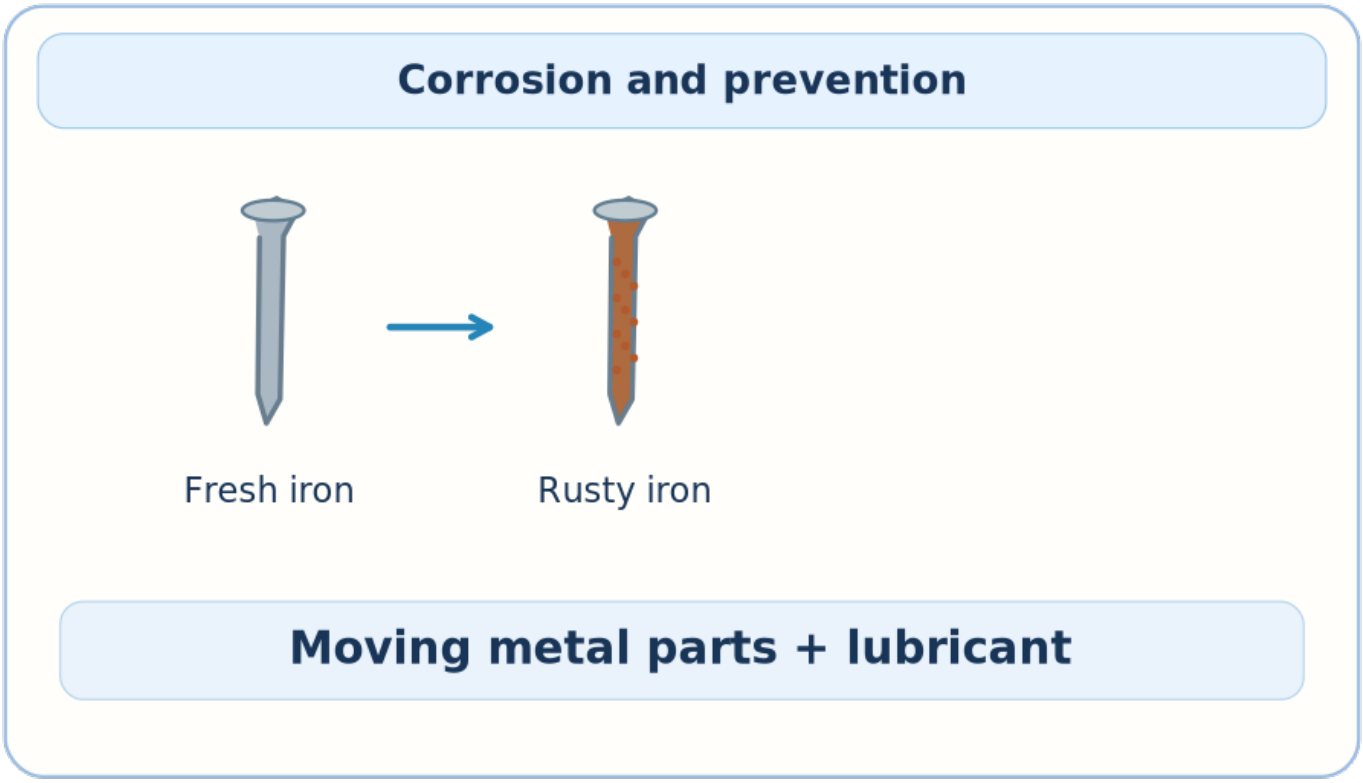
Q6. What is rancidity?



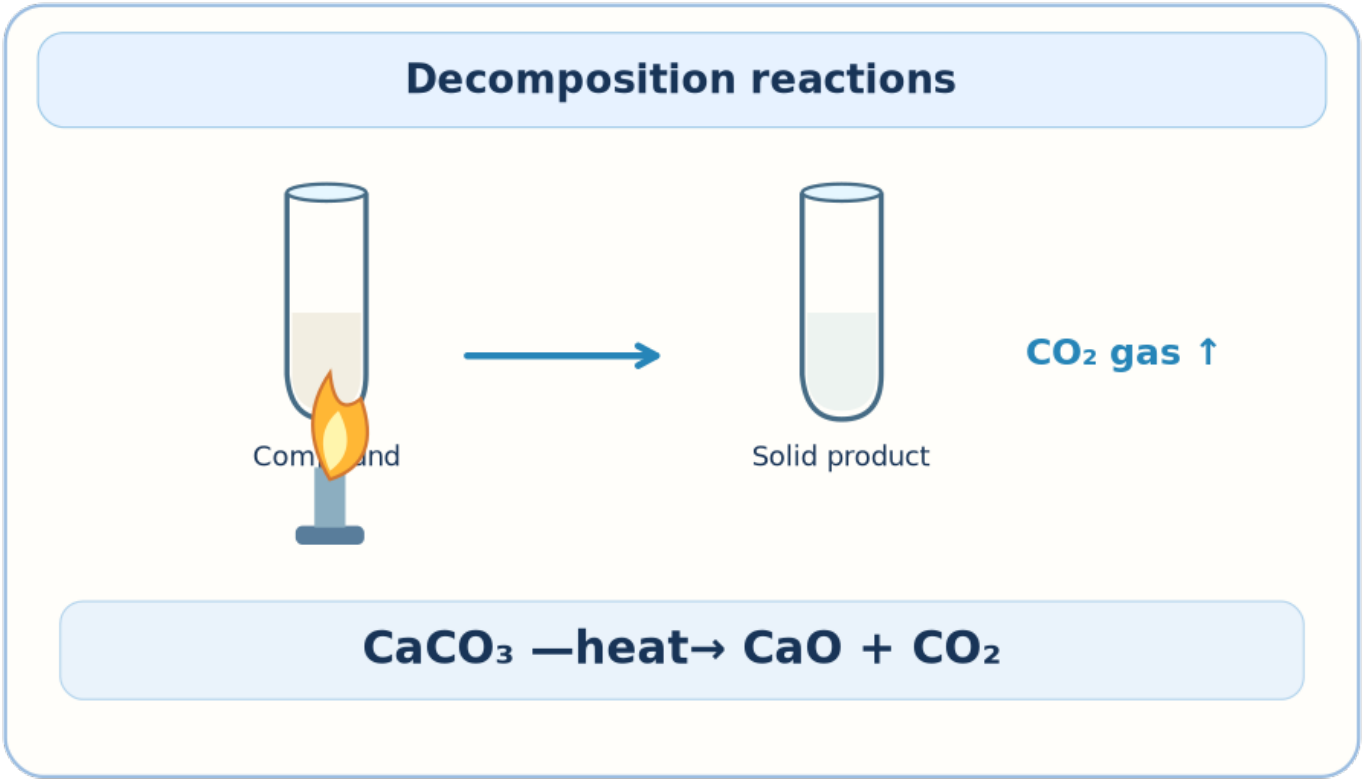
Q7. State observations that indicate a chemical reaction.



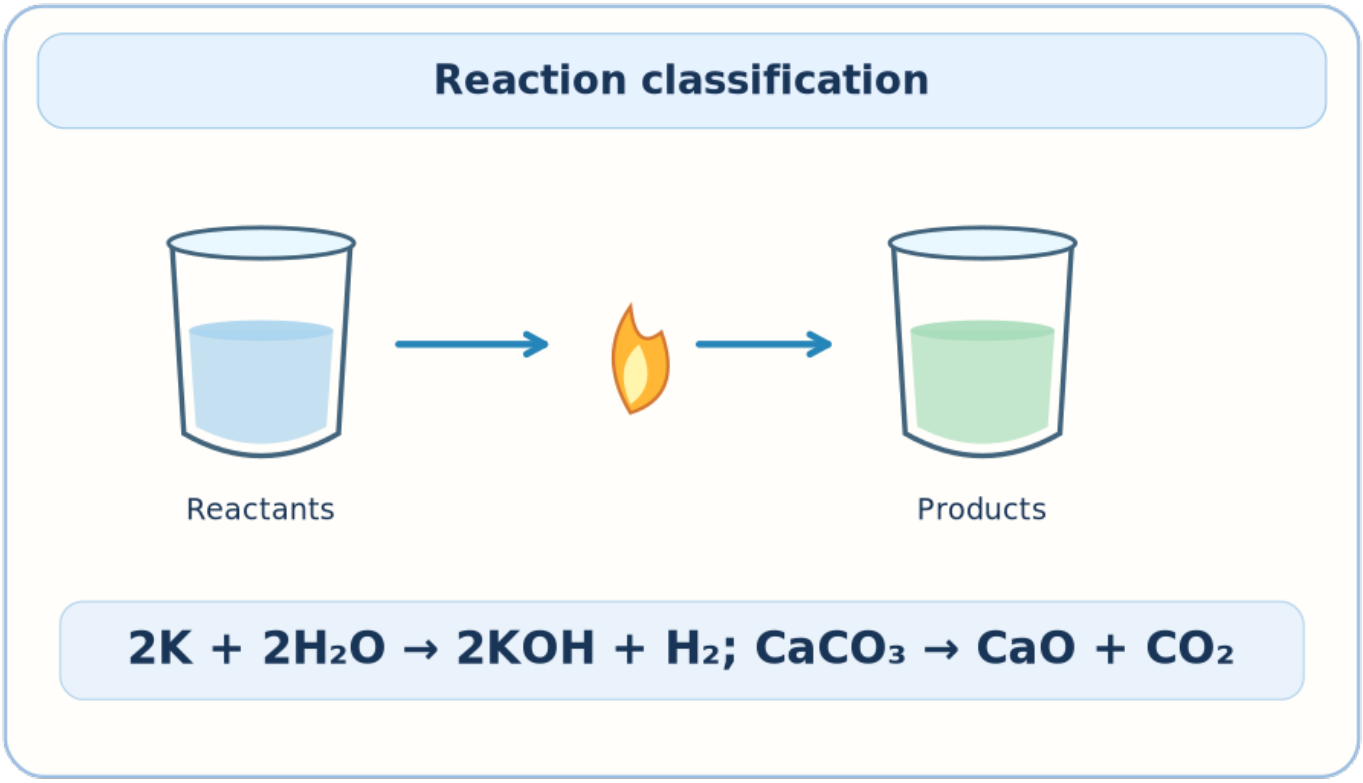
Q8. Why do we apply oil or grease on machines?



Q9. Define a decomposition reaction with an example.



Q10. Identify the reaction types: potassium with water and calcium carbonate on heating.



Q11. Why is rusting considered a chemical change?

Corrosion and prevention



Fresh iron Rusty iron

Iron + oxygen + water → rust

Q12. Why are food packets flushed with nitrogen gas?

Q.12 Why are food packets flushed with nitrogen gas?

Food packet without nitrogen

Air (contains oxygen)



Chips become stale, lose crispness and may develop unpleasant smell due to oxidation of oils.

Food packet flushed with nitrogen

Nitrogen gas (N_2)



Fresh and crispy for a longer time because nitrogen prevents oxidation of oils and fats.

Why nitrogen is used?

1. Nitrogen is an inert gas (does not react with food components).
2. It prevents oxidation of oils and fats present in chips.
3. It prevents rancidity (spoilage) and keeps the chips fresh and crispy.
4. Thus, it increases the shelf life of food.

What is rancidity?

- Rancidity is the oxidation of oils and fats in food, which causes unpleasant smell and taste.

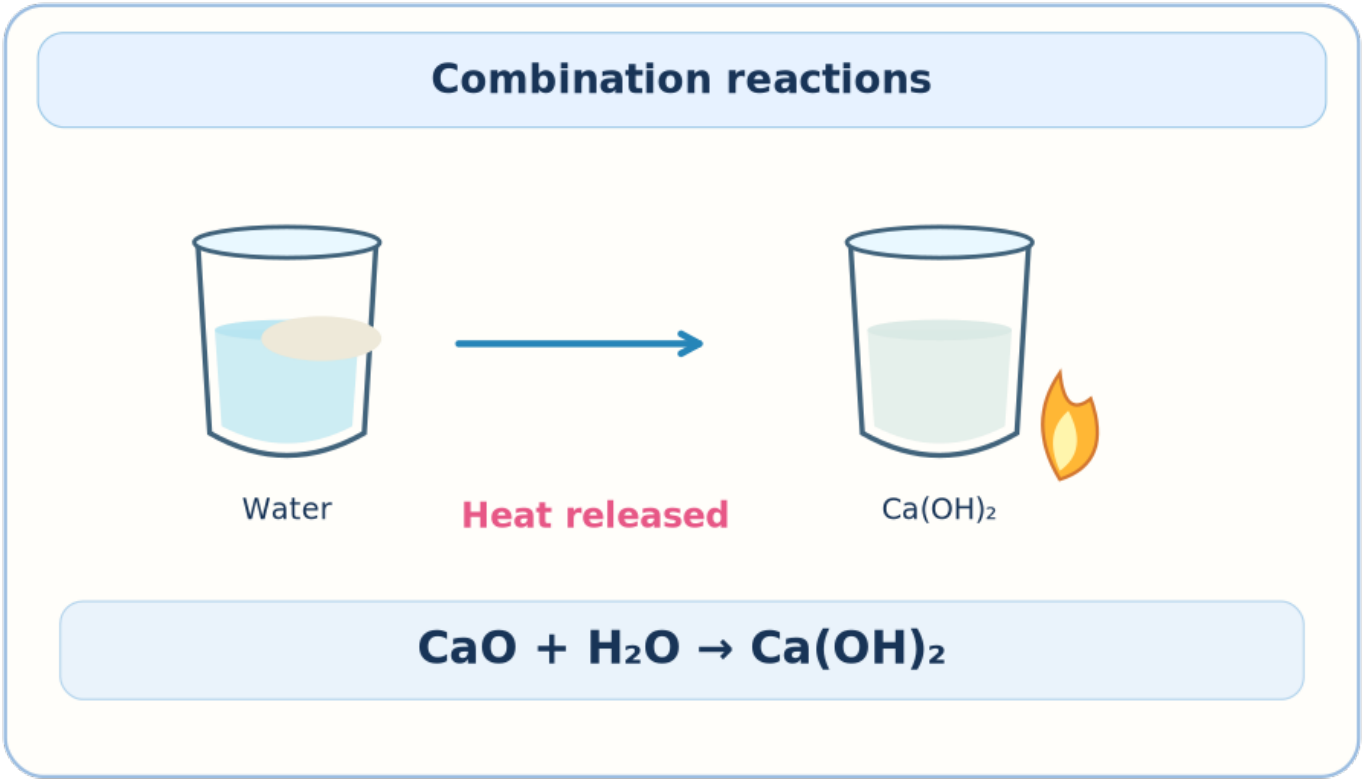


Rancid chip (unpleasant smell)

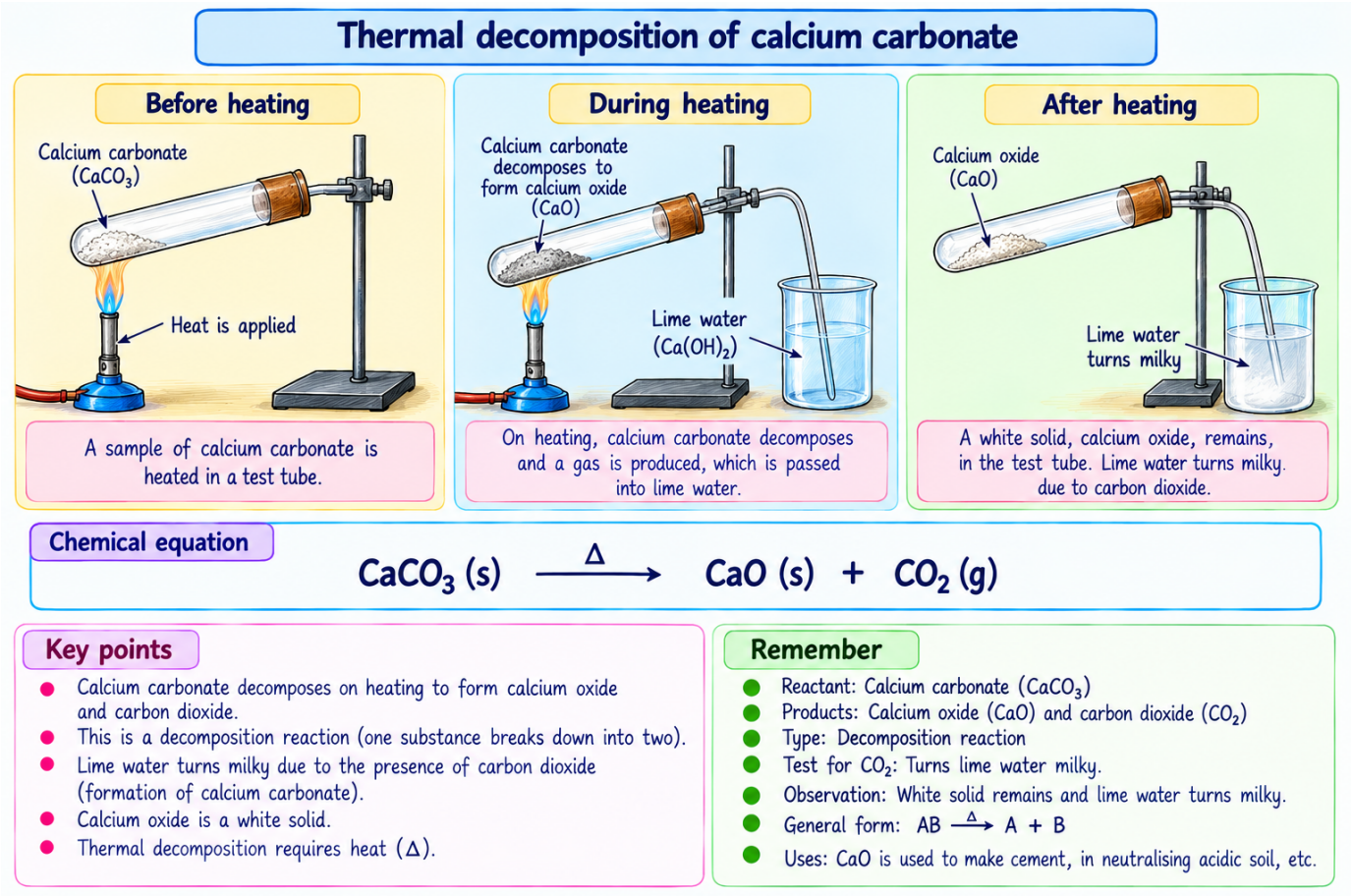
Key takeaways

- Nitrogen is an inert gas.
- It prevents oxidation and rancidity of food.
- It keeps food fresh for a longer time.
- This is why many food packets are flushed with nitrogen gas before sealing.

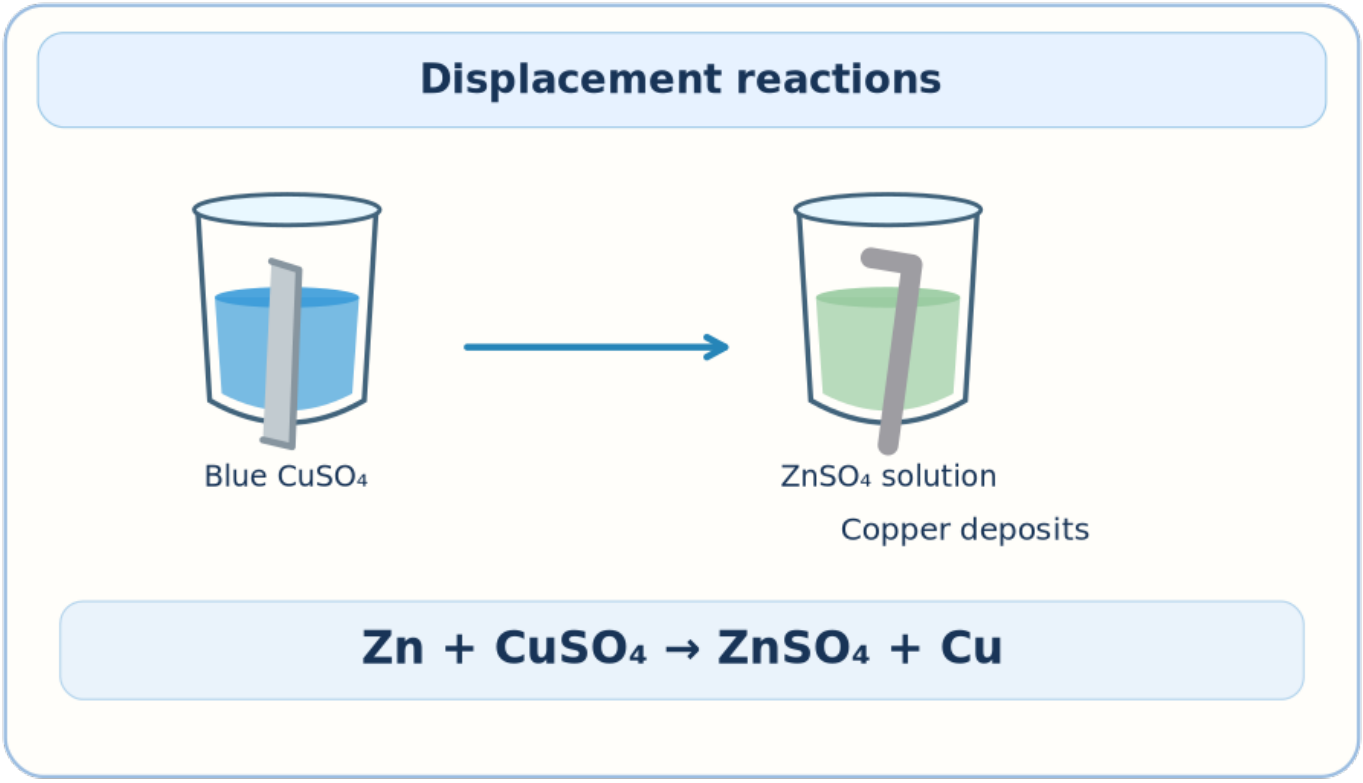
Q13. Define combination reaction and give an example.



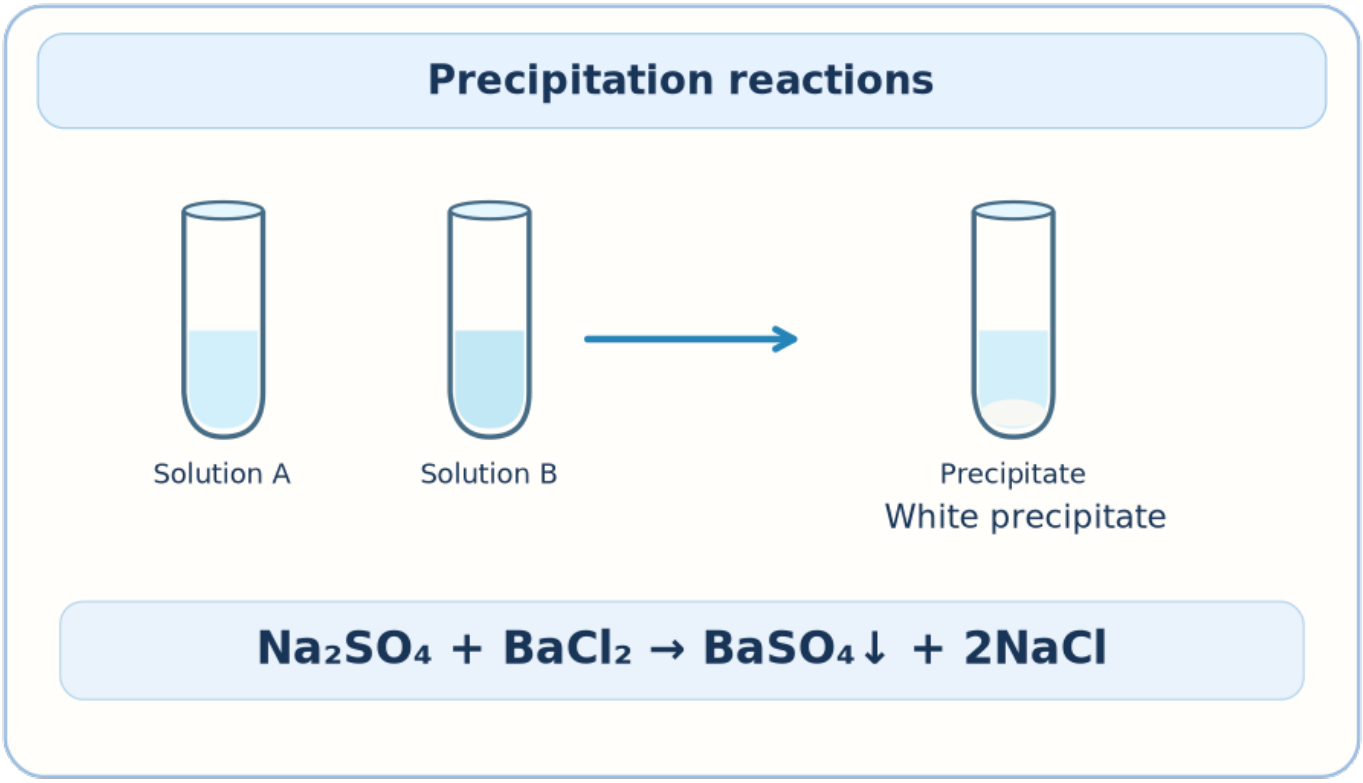
Q14. Give an example of decomposition caused by heat.



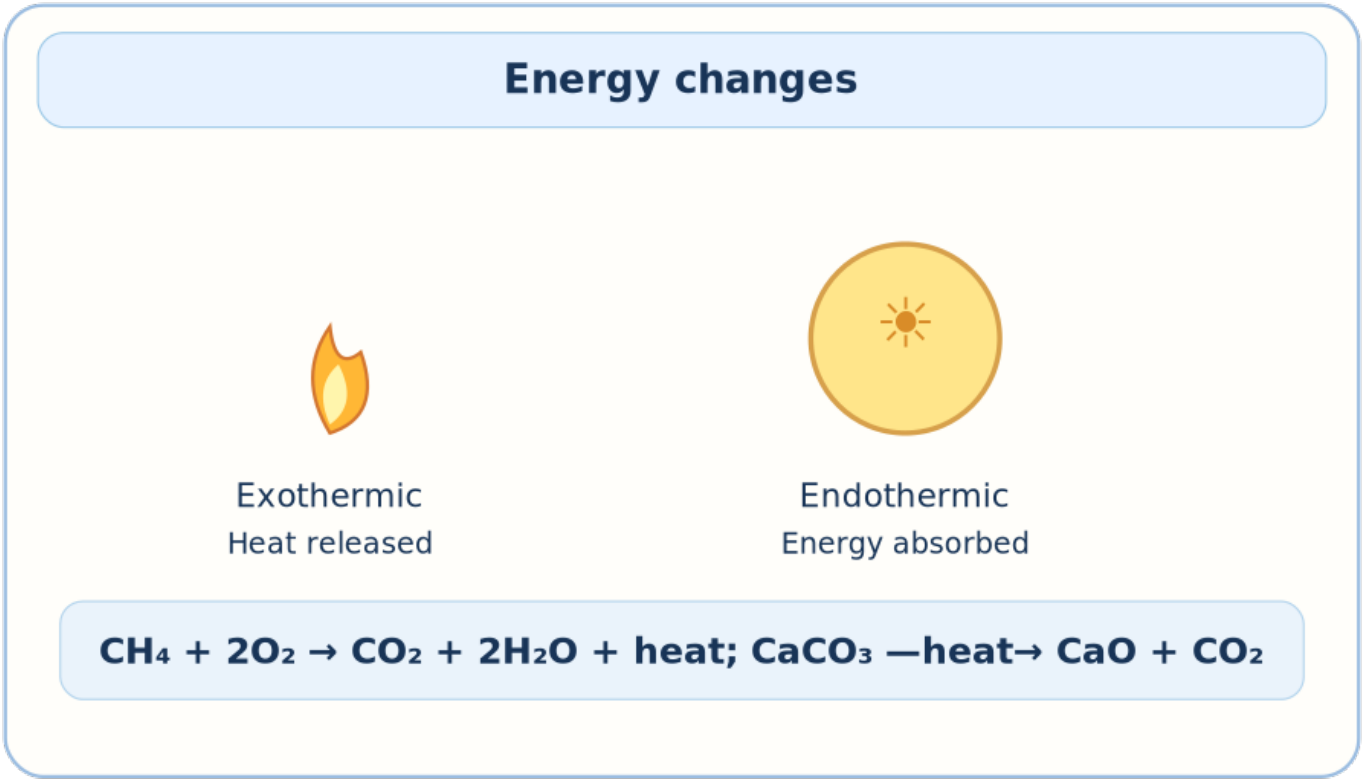
Q15. Give an example of displacement reaction.



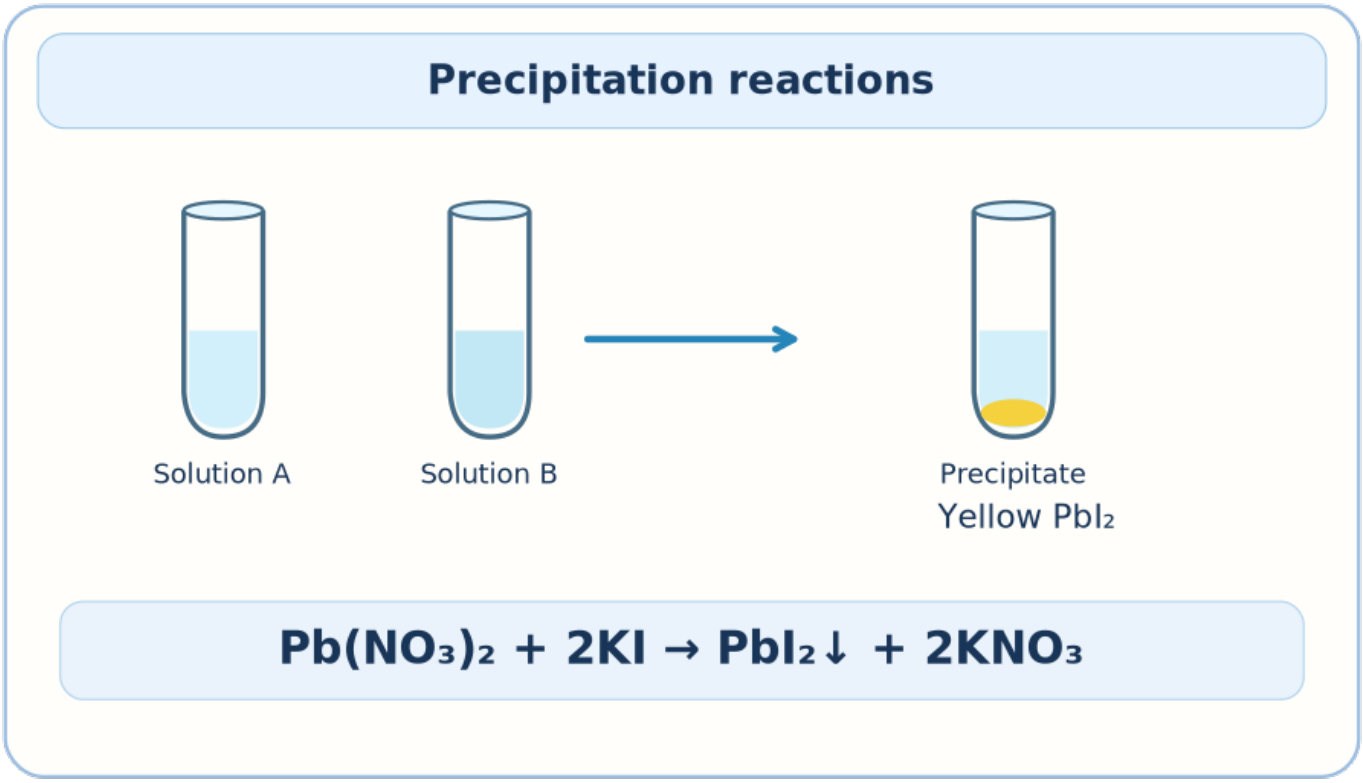
Q16. Give an example of double displacement reaction.



Q17. Differentiate exothermic and endothermic reactions.

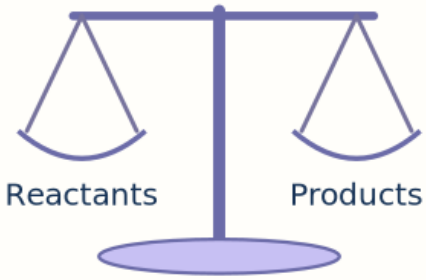


Q18. What happens when lead nitrate and potassium iodide solutions are mixed? Write the equation.



Q19. Why do we balance chemical equations?

Balancing equations



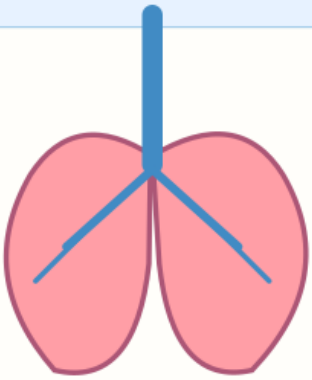
Atoms are conserved

Count each element

$$2Mg + O_2 \rightarrow 2MgO$$

Q20. What is respiration? Write its balanced equation.

Respiration



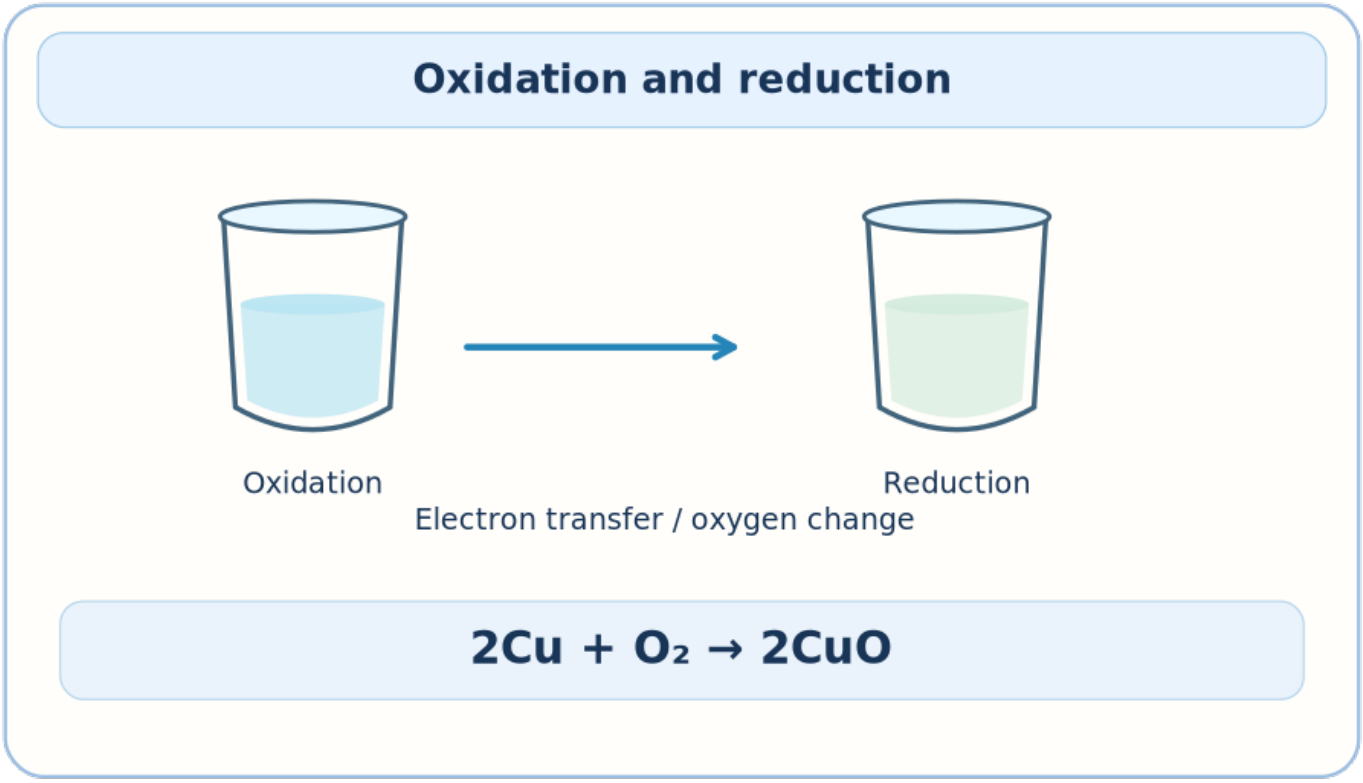
Glucose + oxygen

↓

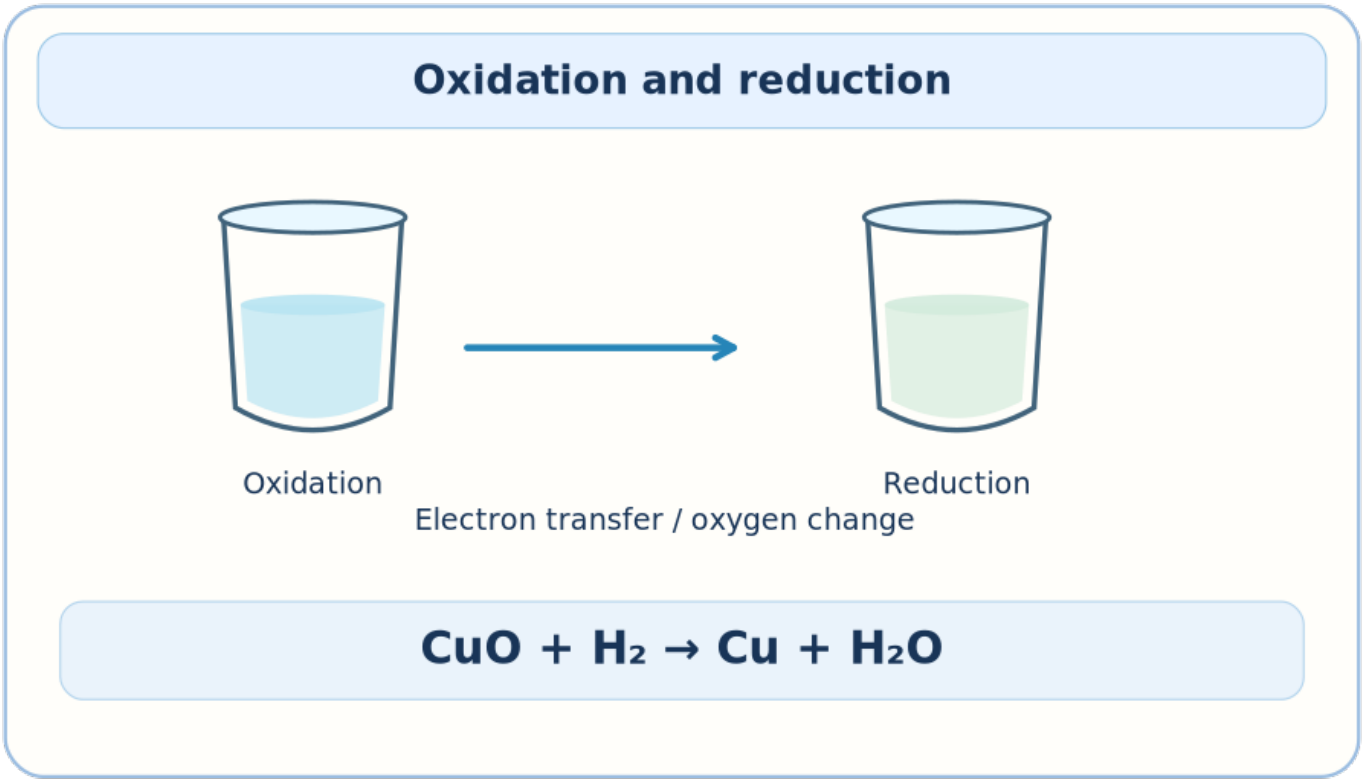
Energy released

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$$

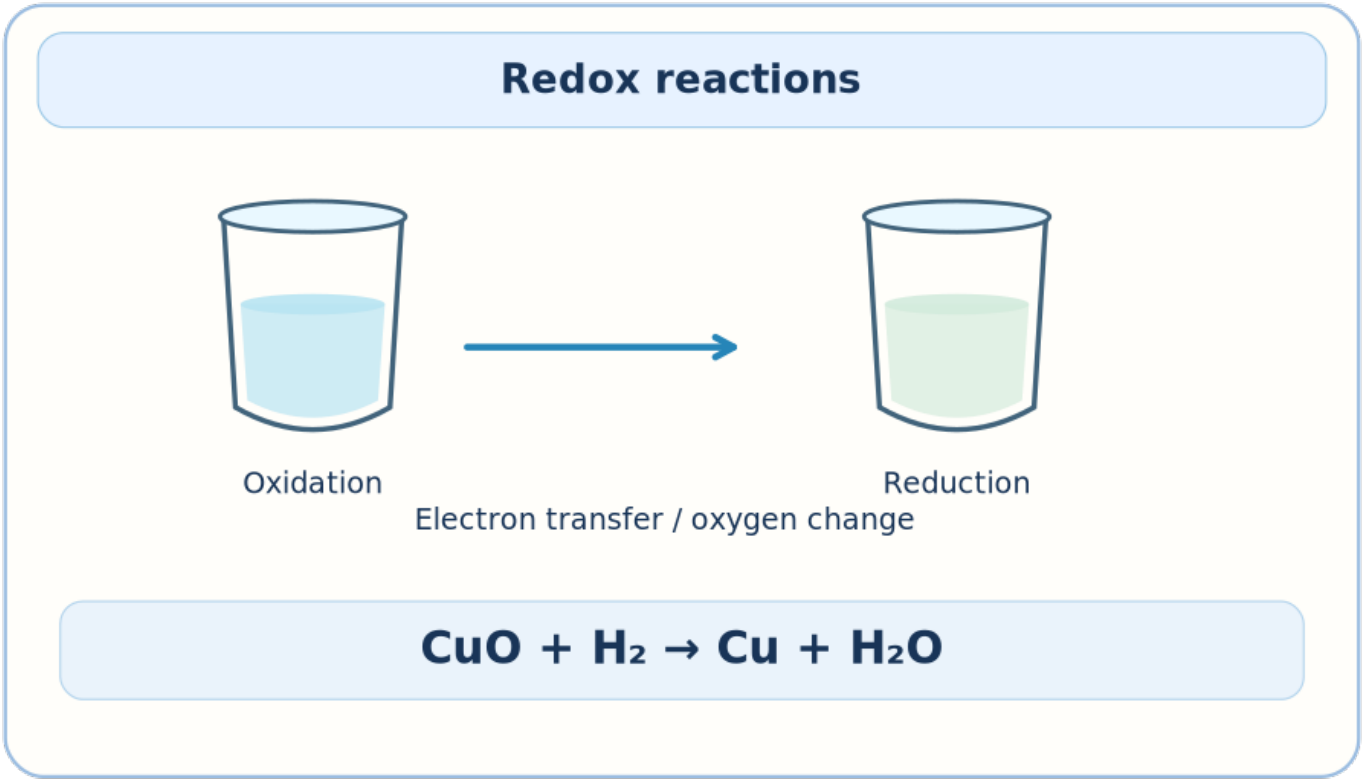
Q21. What is oxidation? Give an example.



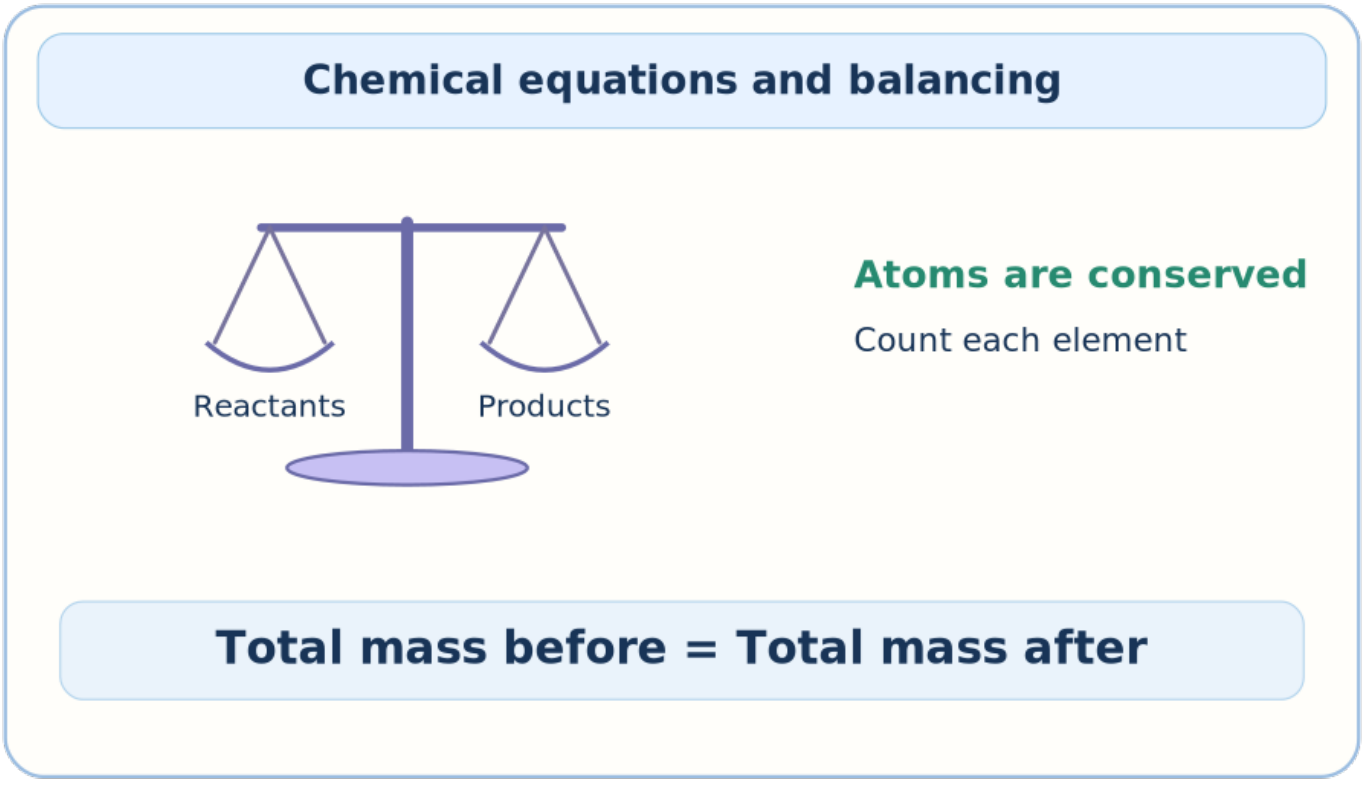
Q22. What is reduction? Give an example.



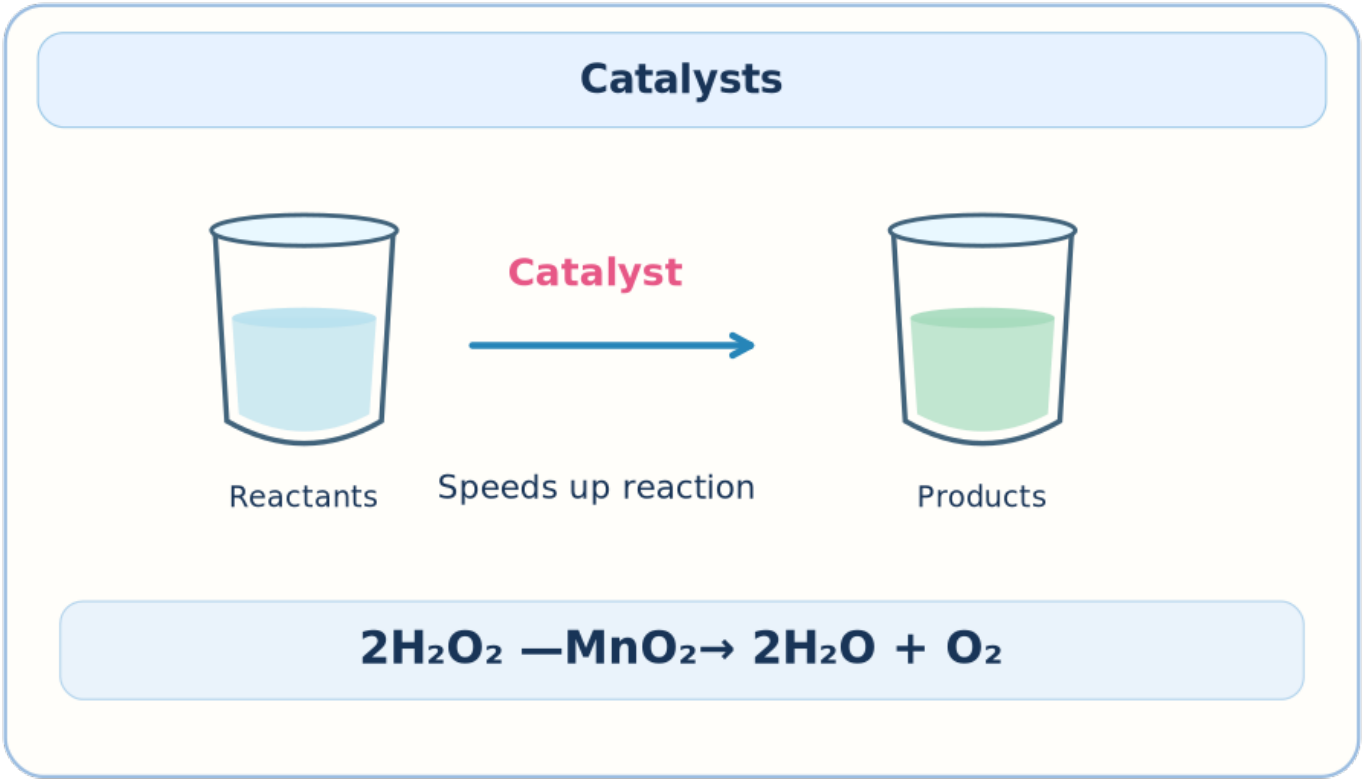
Q23. What is a redox reaction? Give an example.



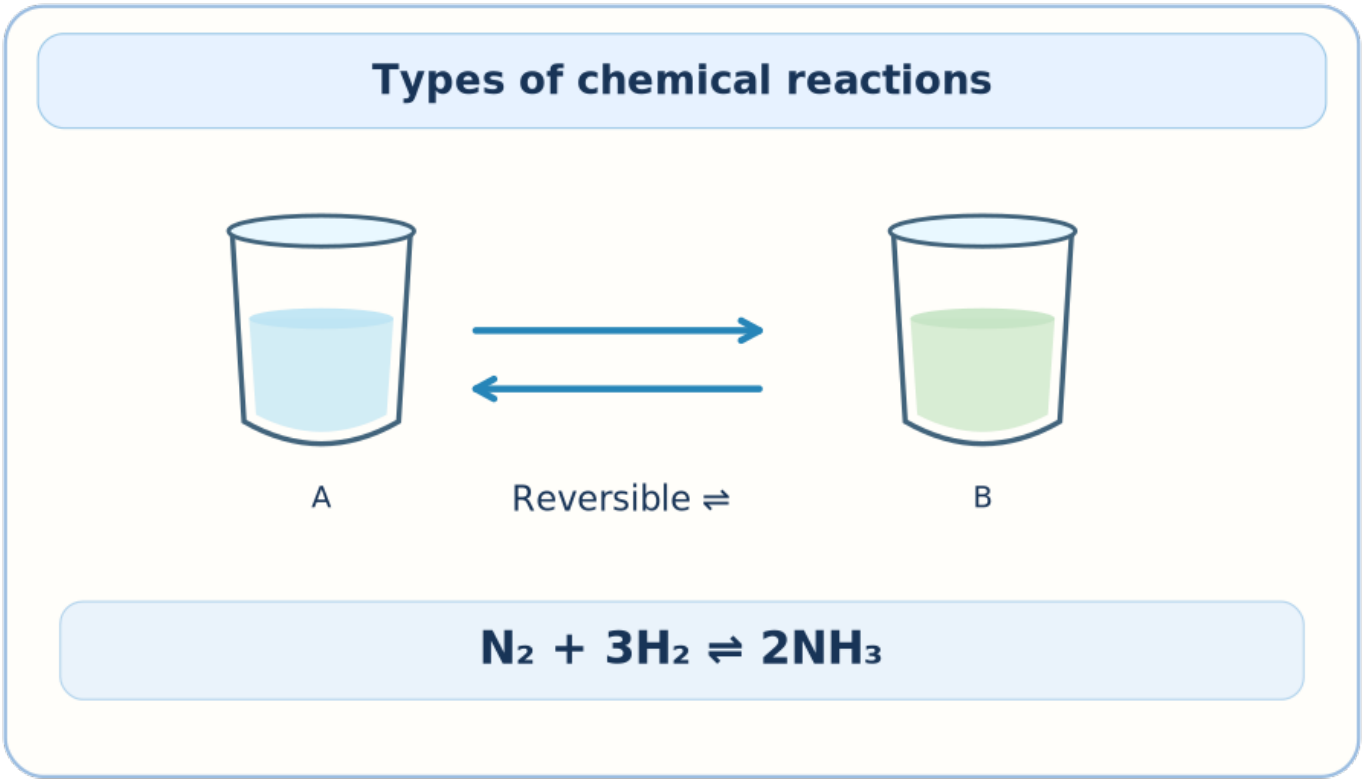
Q24. State the law of conservation of mass.



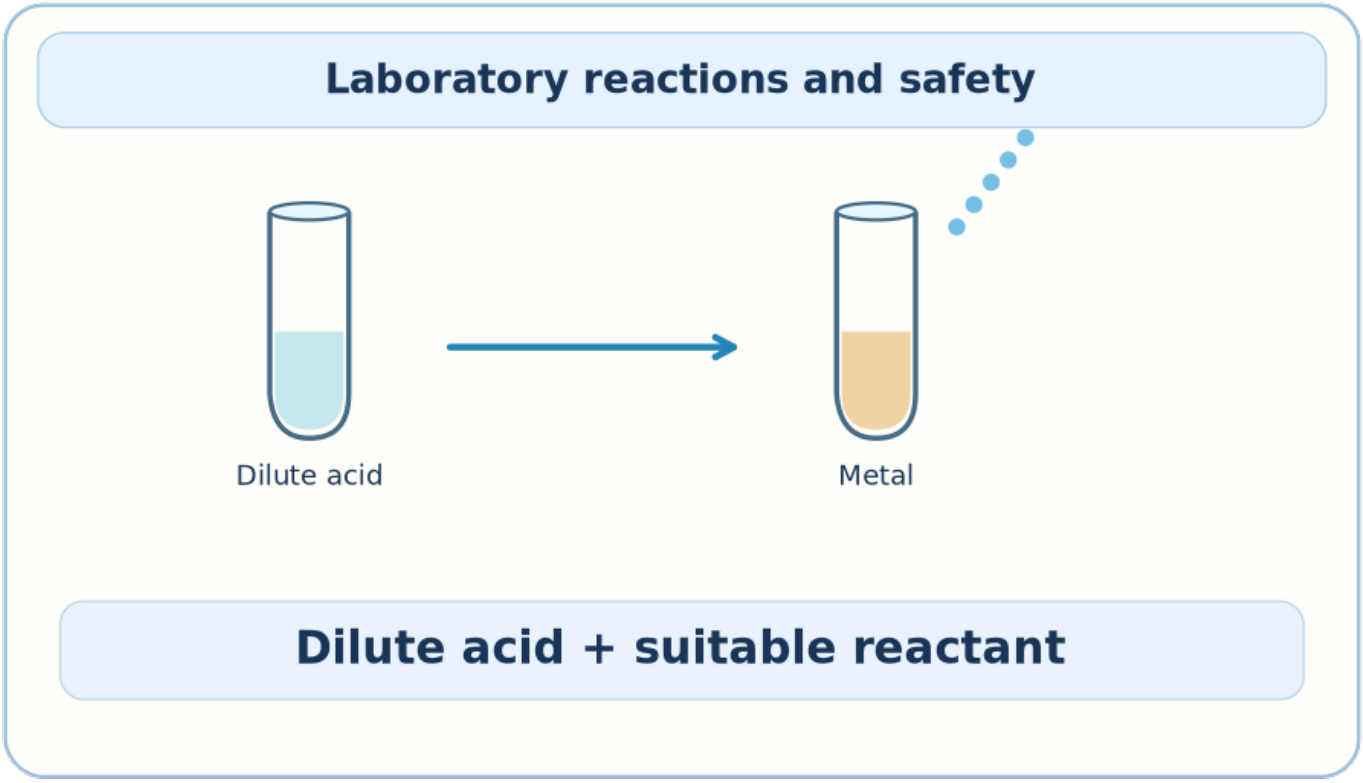
Q25. What is the role of a catalyst in a chemical reaction?



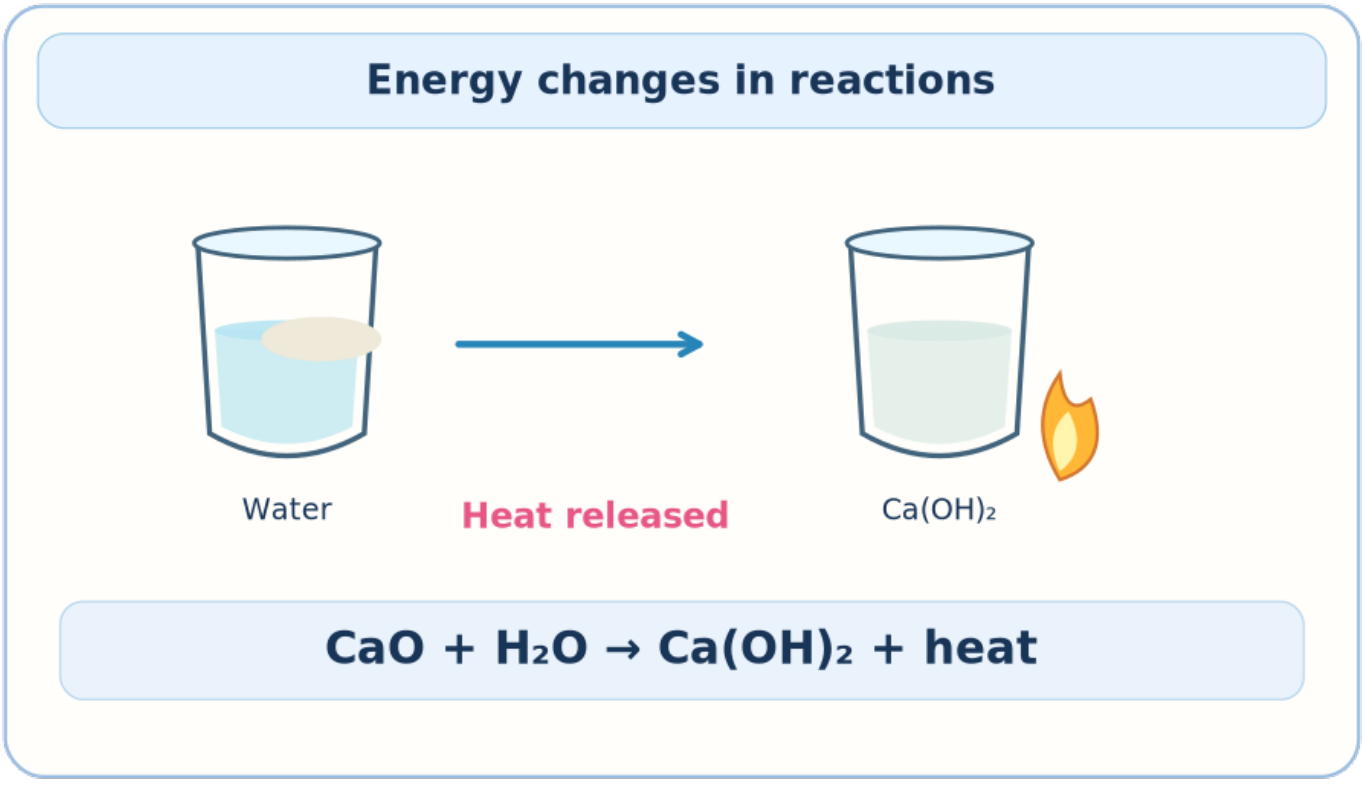
Q26. Differentiate reversible and irreversible reactions.



Q27. Why do we generally use dilute acids in chemical reactions?



Q28. What happens when calcium oxide is added to water? Write equation.



Q29. What is corrosion? How can it be prevented?

Corrosion prevention



Fresh iron Rusty iron

Iron + oxygen + moisture → rust

Q30. Why are thermite reactions used in welding railway tracks?

Energy changes in reactions

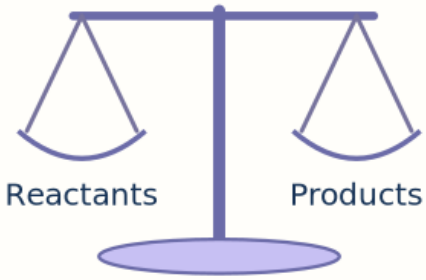


Thermite welding

$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + heat$

Q31. Write balanced chemical equations for the given word equations.

Balancing equations

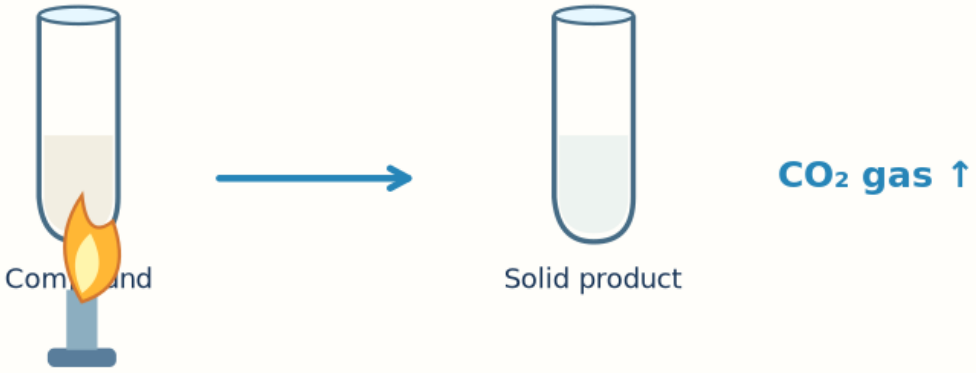


Atoms are conserved

Count each element

Q32. Differentiate combination and decomposition reactions.

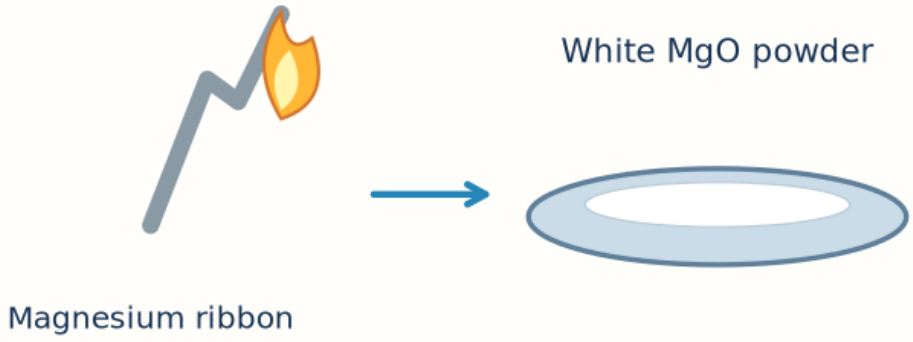
Reaction classification



$CaO + H_2O \rightarrow Ca(OH)_2$; $CaCO_3 \rightarrow CaO + CO_2$

Q33. What is the role of oxygen in burning magnesium?

Combination reactions



$$2Mg + O_2 \rightarrow 2MgO$$

Q34. What happens when sodium is added to water? Give equation.

Displacement reactions

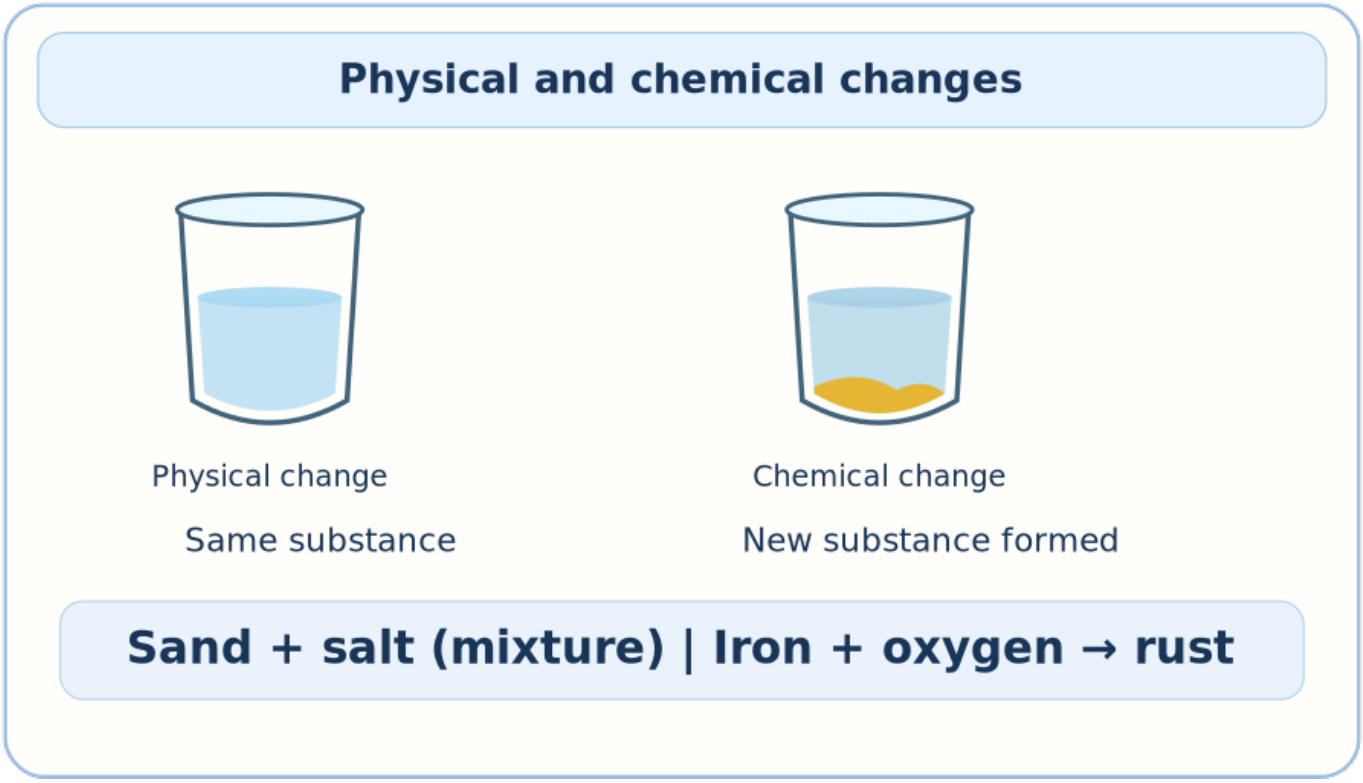


Hydrogen gas ↑

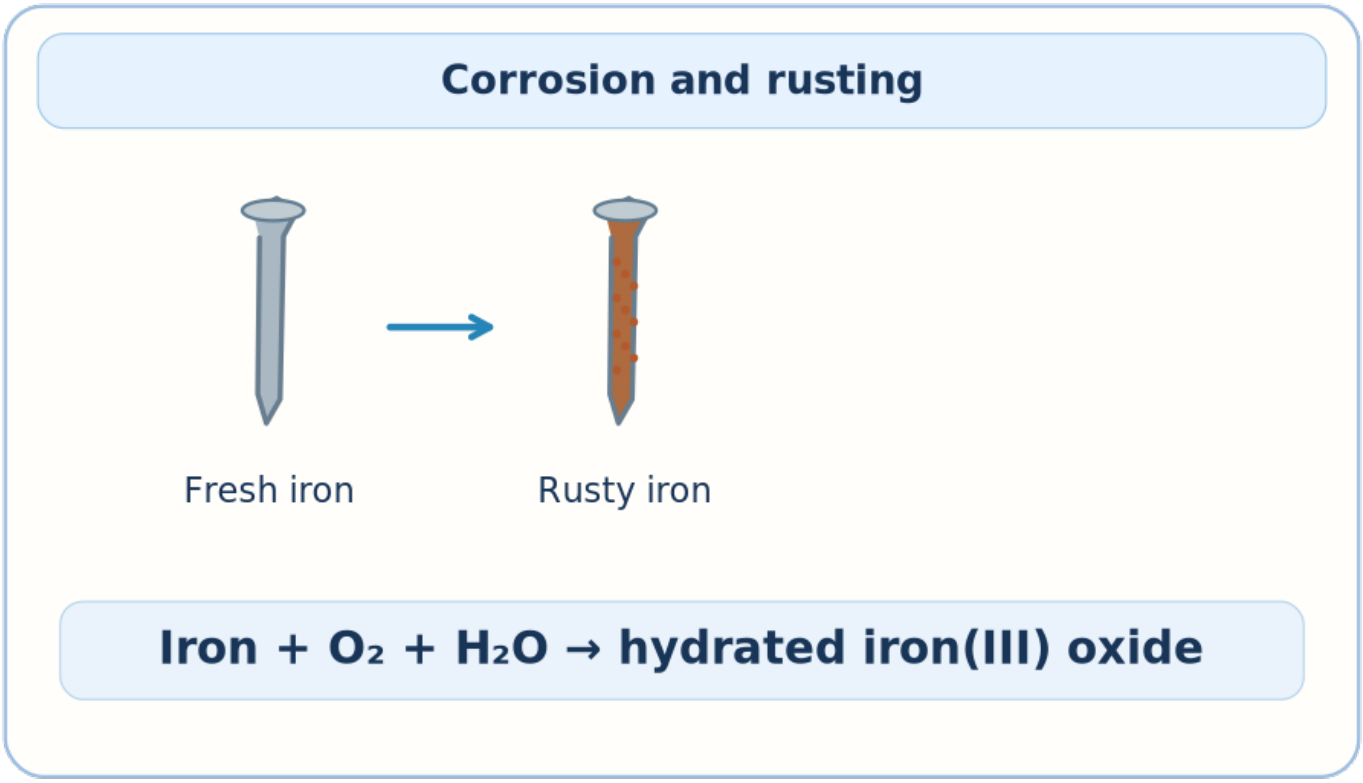
Heat released

$$2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$$

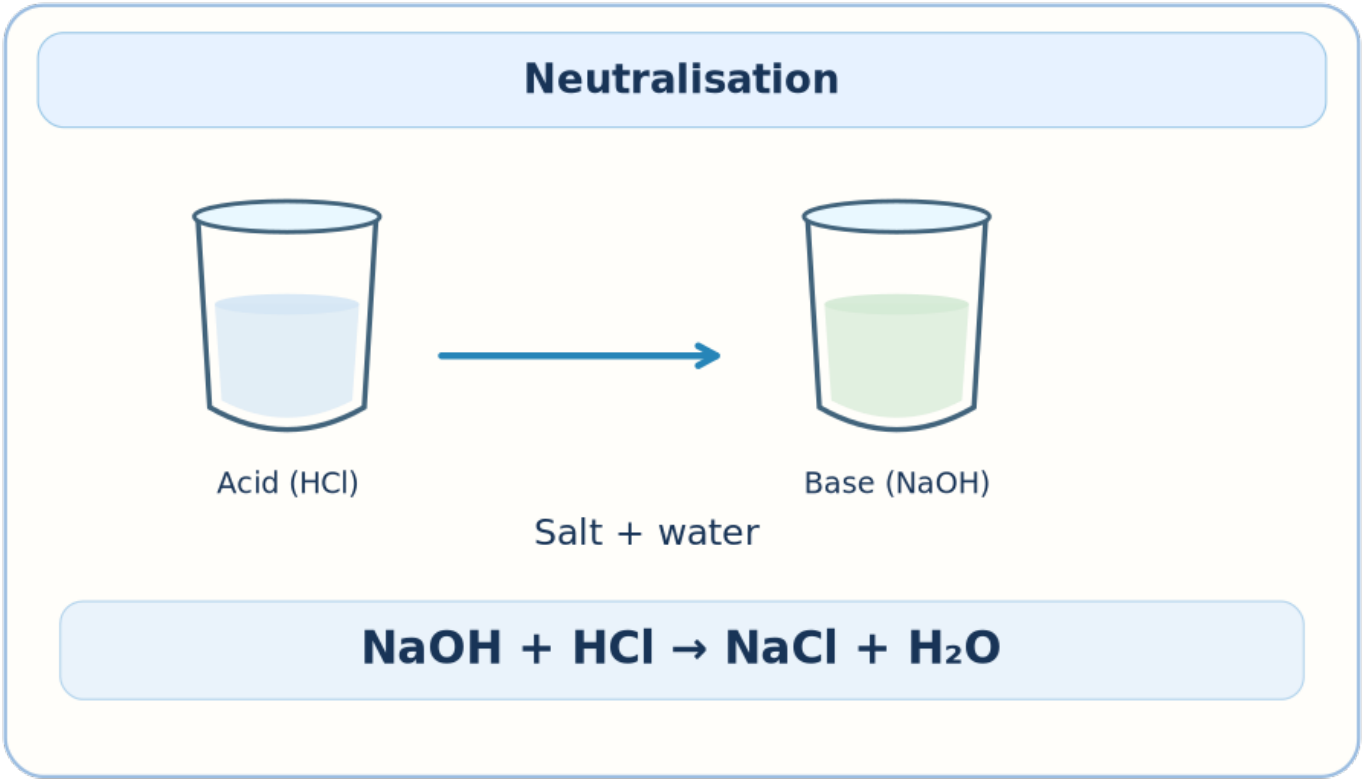
Q35. Differentiate physical mixture and chemical reaction.



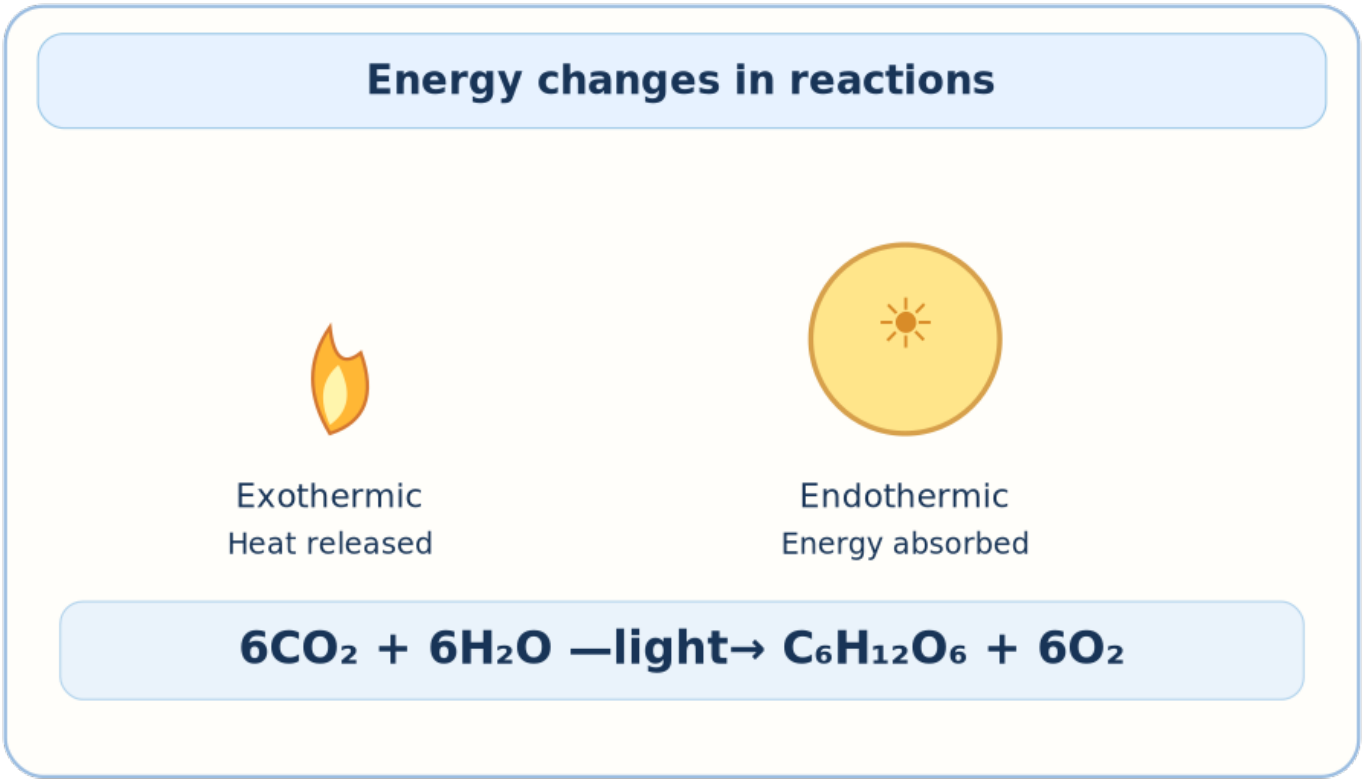
Q36. Why is rusting of iron considered a slow chemical reaction?



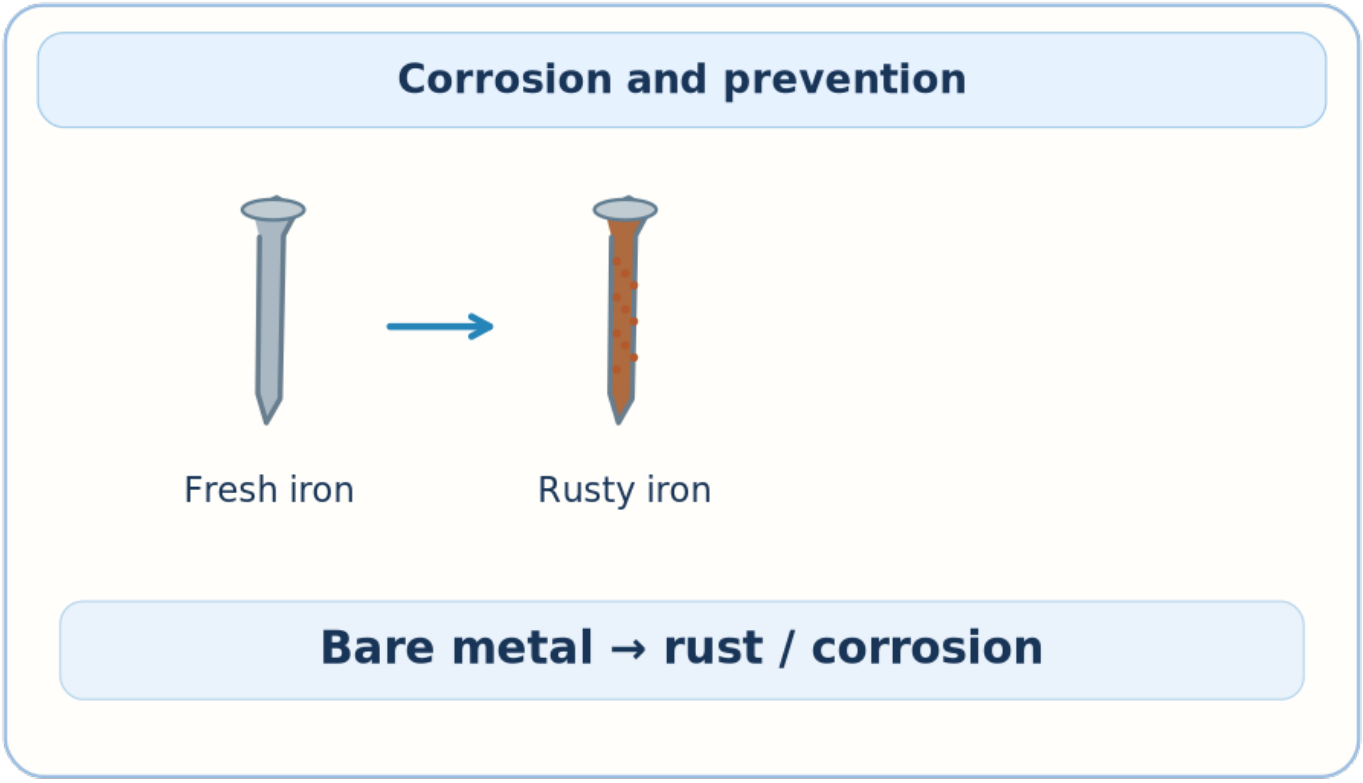
Q37. Write balanced equation for neutralisation of sodium hydroxide with hydrochloric acid.



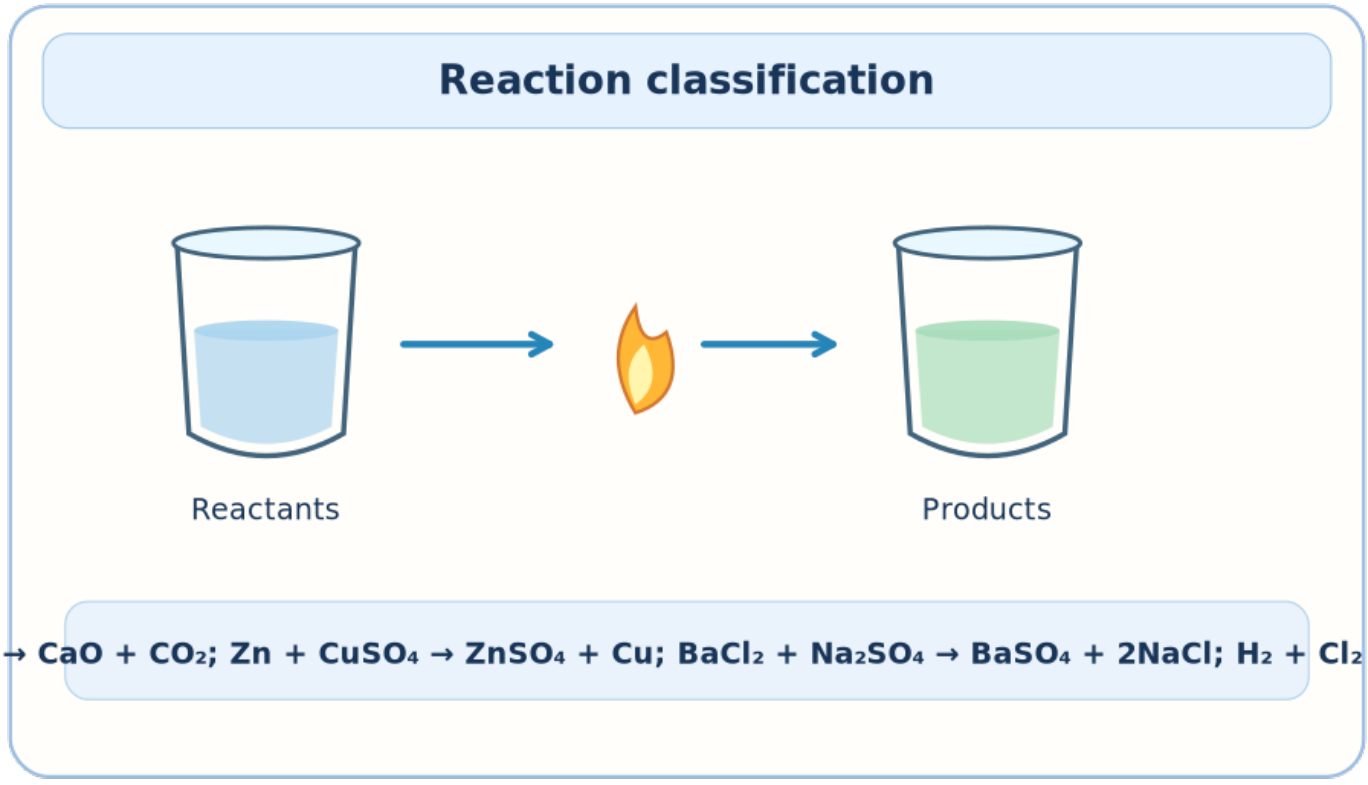
Q38. Why is photosynthesis an endothermic process?



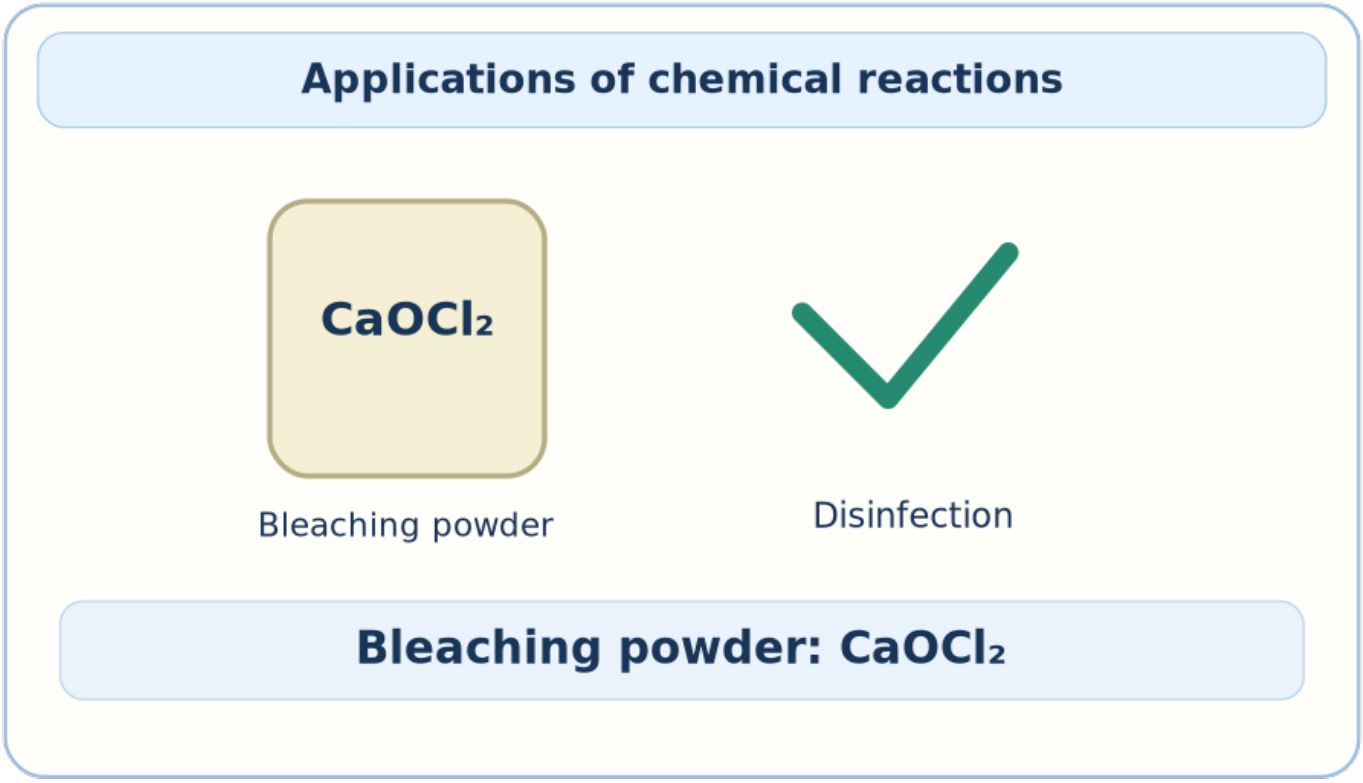
Q39. What are the effects of corrosion?



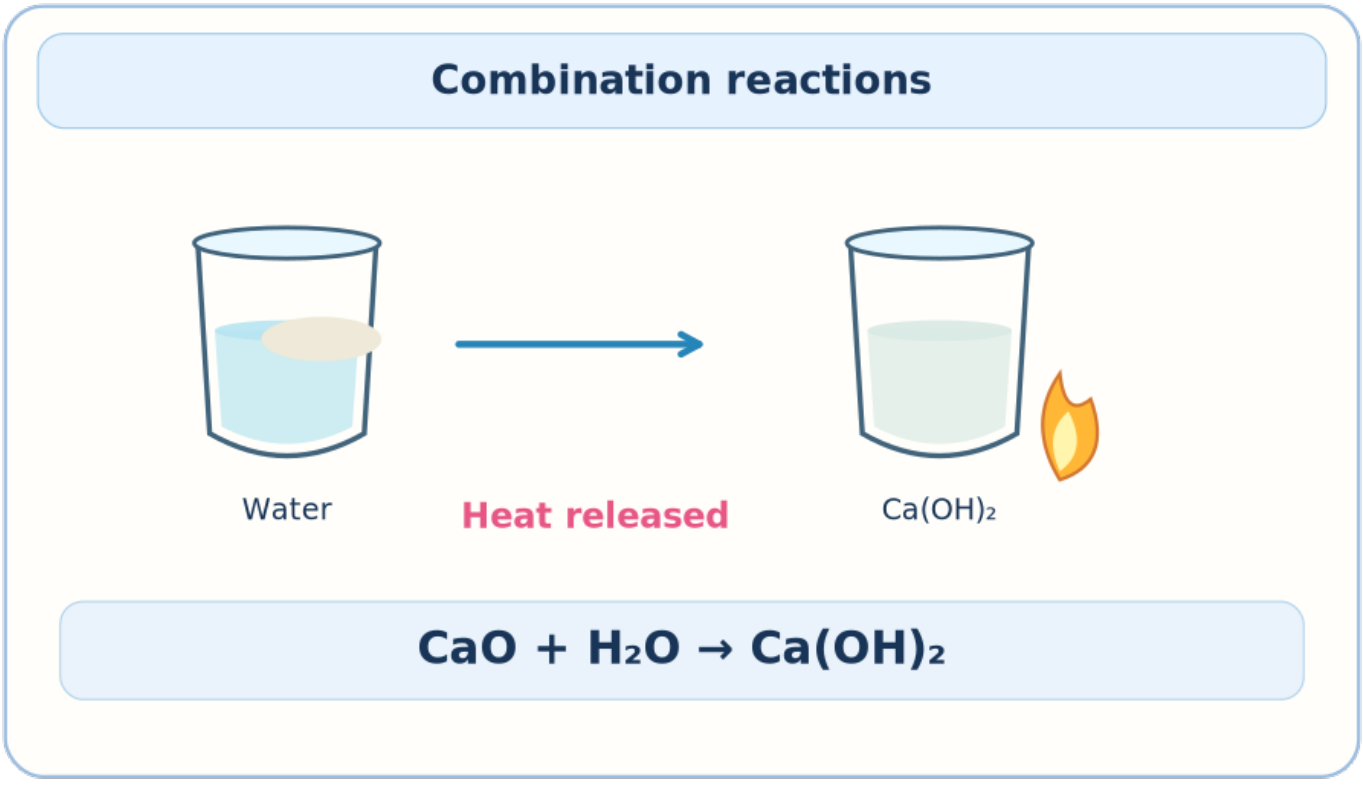
Q40. Classify four reactions and write balanced equations.



Q41. What is bleaching powder? Give formula and one use.



Q42. Why is calcium oxide called a basic oxide? Write reaction with water.



Q43. What happens when steam is passed over red-hot iron?

Oxidation and reduction



Oxidation

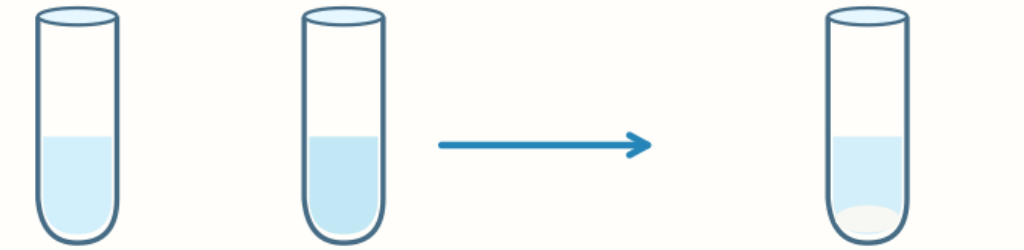
Reduction

Electron transfer / oxygen change

$$3Fe + 4H_2O(g) \rightarrow Fe_3O_4 + 4H_2$$

Q44. Write a balanced equation for barium hydroxide and aluminium sulphate.

Precipitation reactions



Solution A

Solution B

Precipitate

White precipitate

$$3Ba(OH)_2 + Al_2(SO_4)_3 \rightarrow 3BaSO_4\downarrow + 2Al(OH)_3\downarrow$$

Q45. What happens when iron filings are added to copper sulphate solution?

Displacement reactions



$$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$$

Q46. Write balanced equation for combustion of methane.

Energy changes in reactions



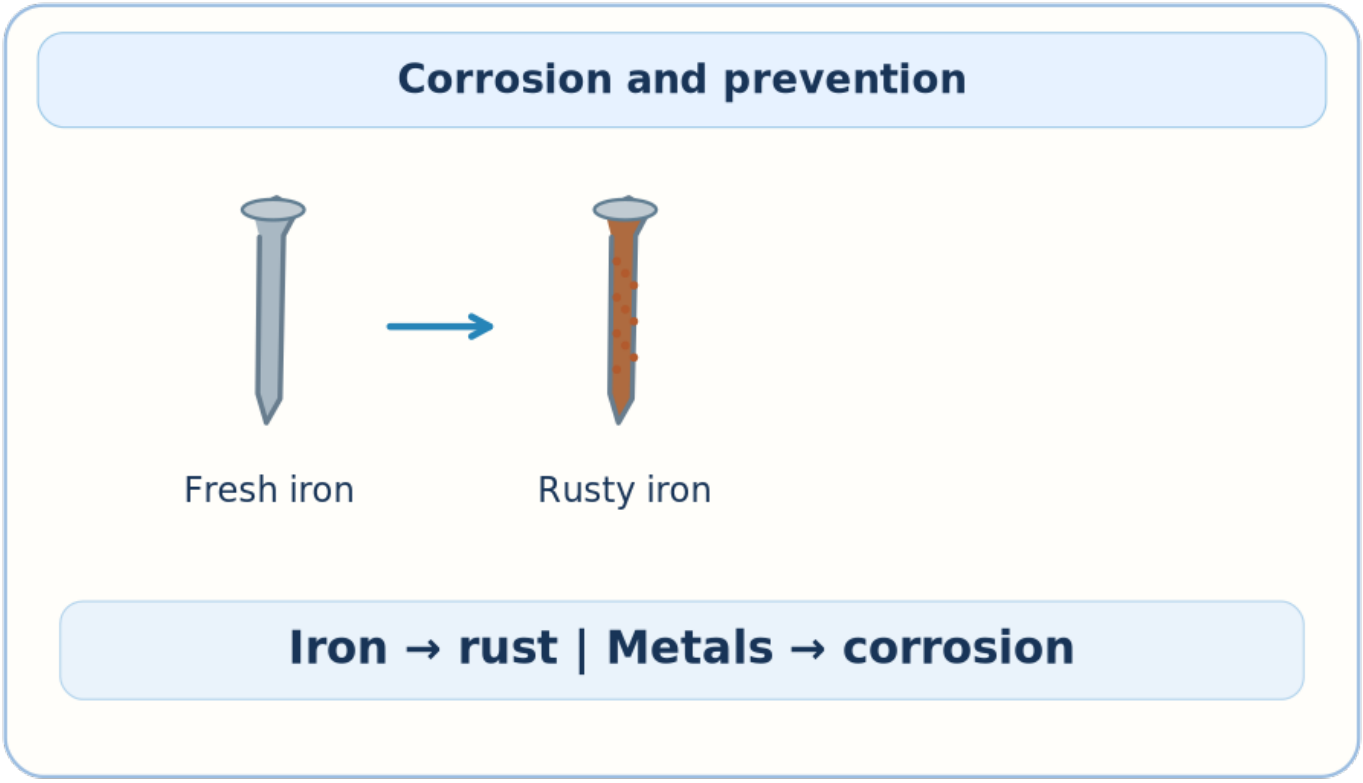
Methane + oxygen

↓

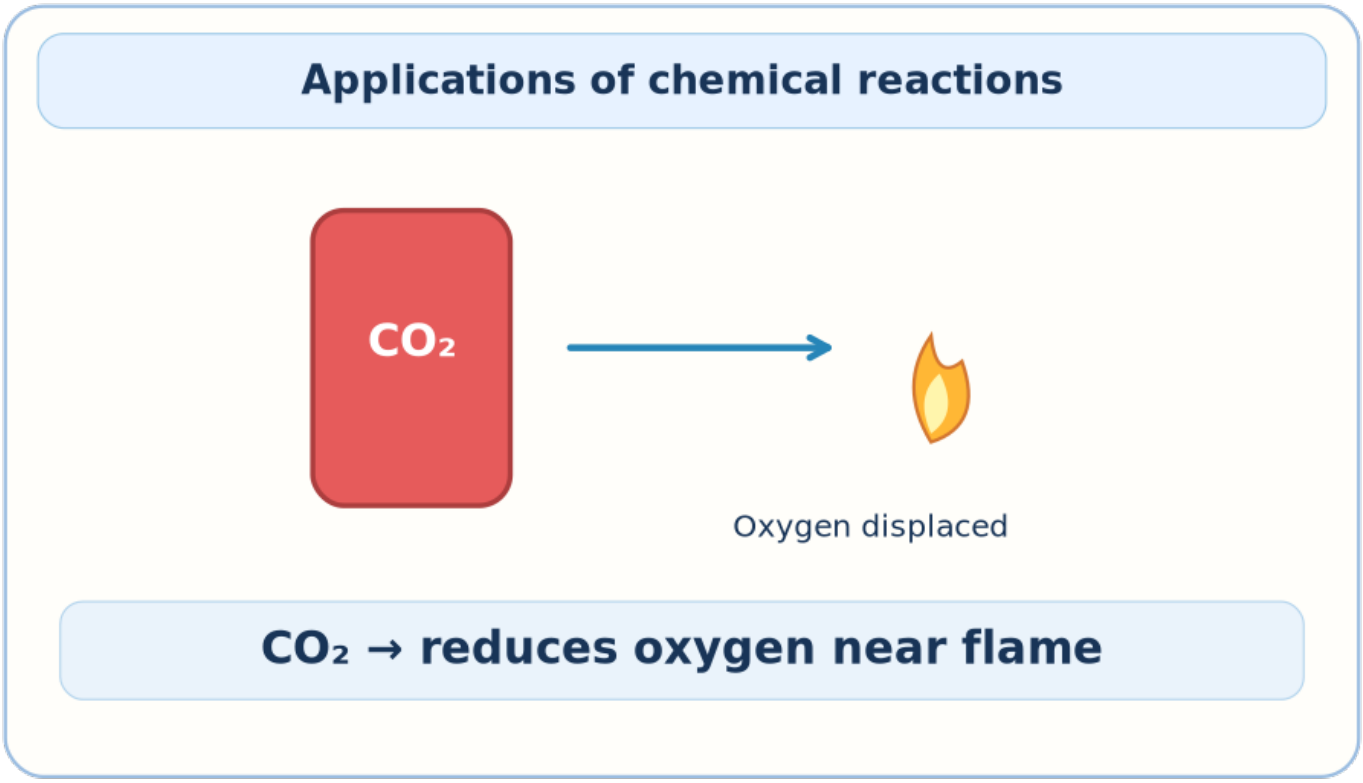
$CO_2 + H_2O$ + heat

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + heat$$

Q47. Differentiate rusting and corrosion.



Q48. What is the role of carbon dioxide in a fire extinguisher?



Q49. Write the balanced equation for reaction between hydrogen and oxygen.

Combination reactions



Hydrogen + oxygen Water

$$2H_2 + O_2 \rightarrow 2H_2O$$

Q50. What happens when dilute acetic acid is added to sodium hydrogen carbonate? Give equation.

Gas evolution reactions



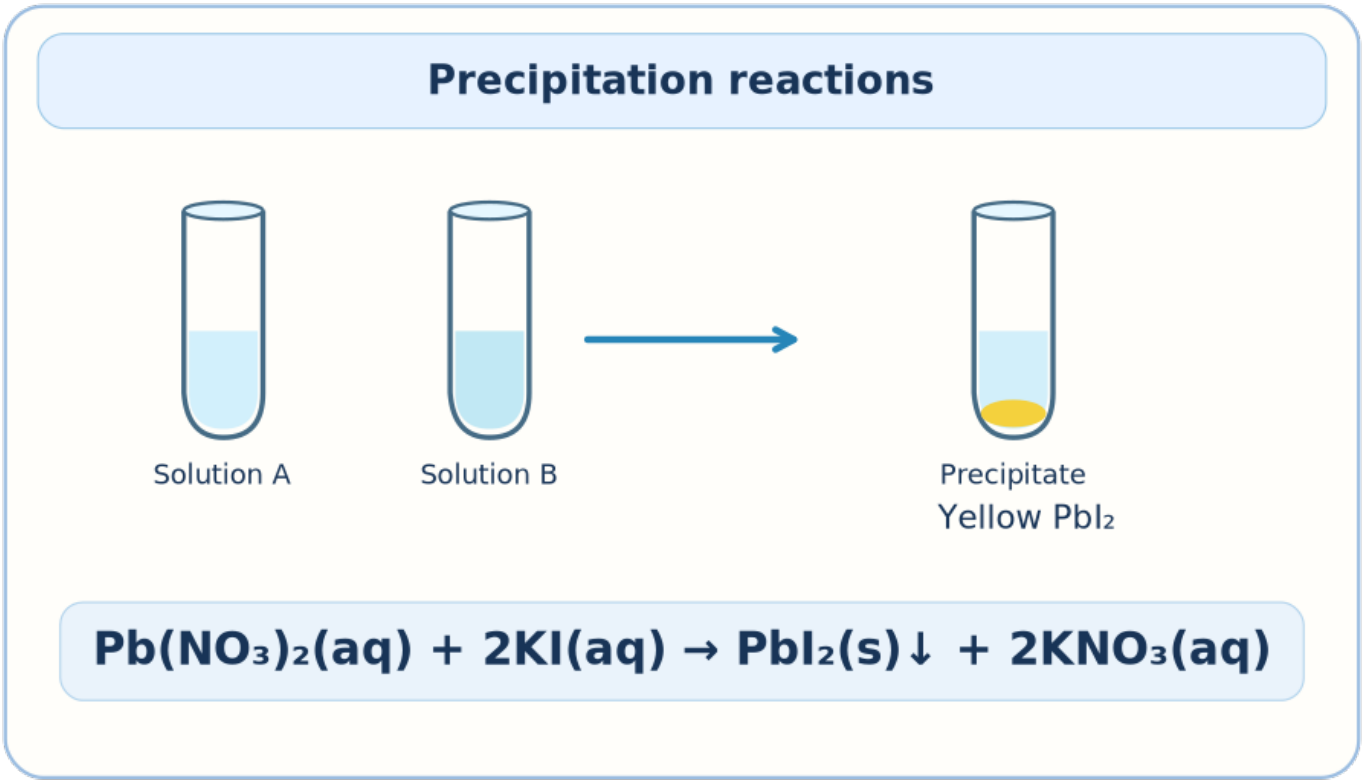
Vinegar

CO_2 bubbles ↑

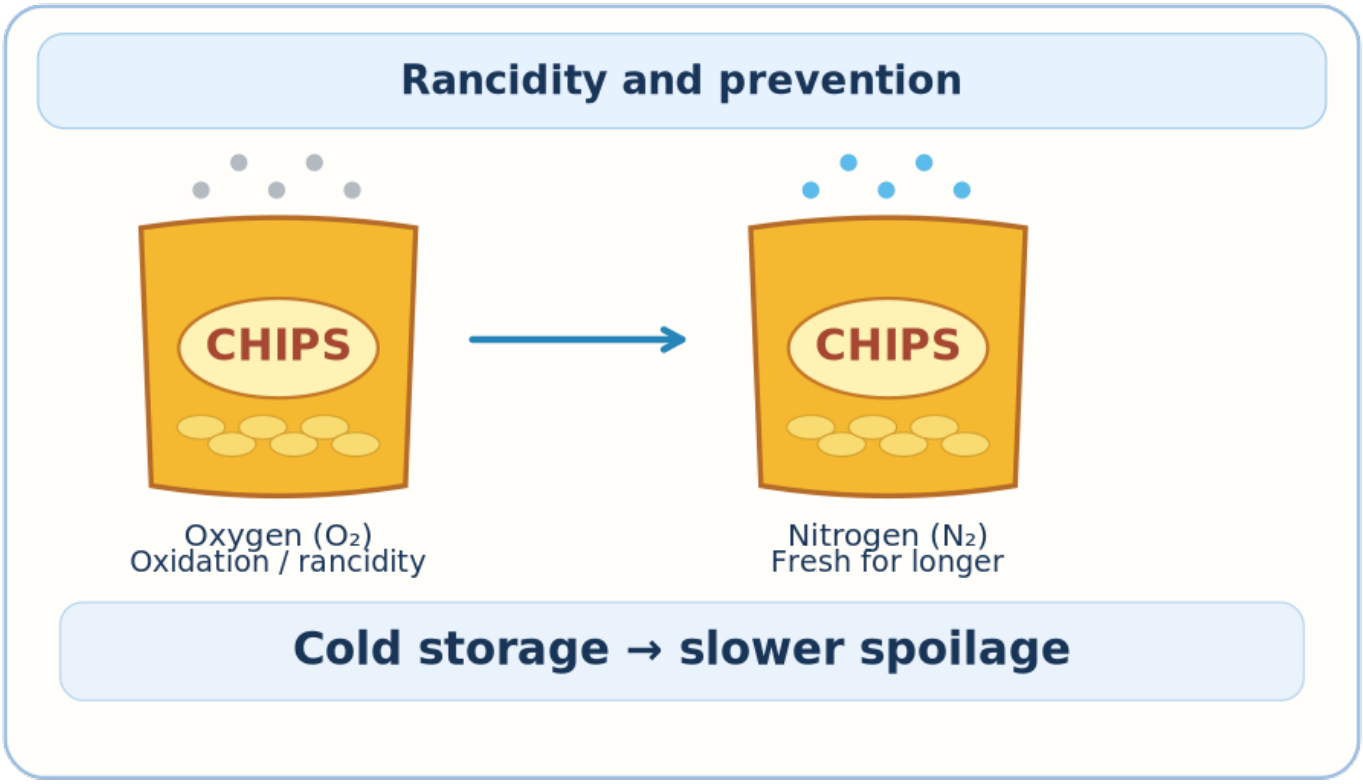
Baking soda reaction

$$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2\uparrow$$

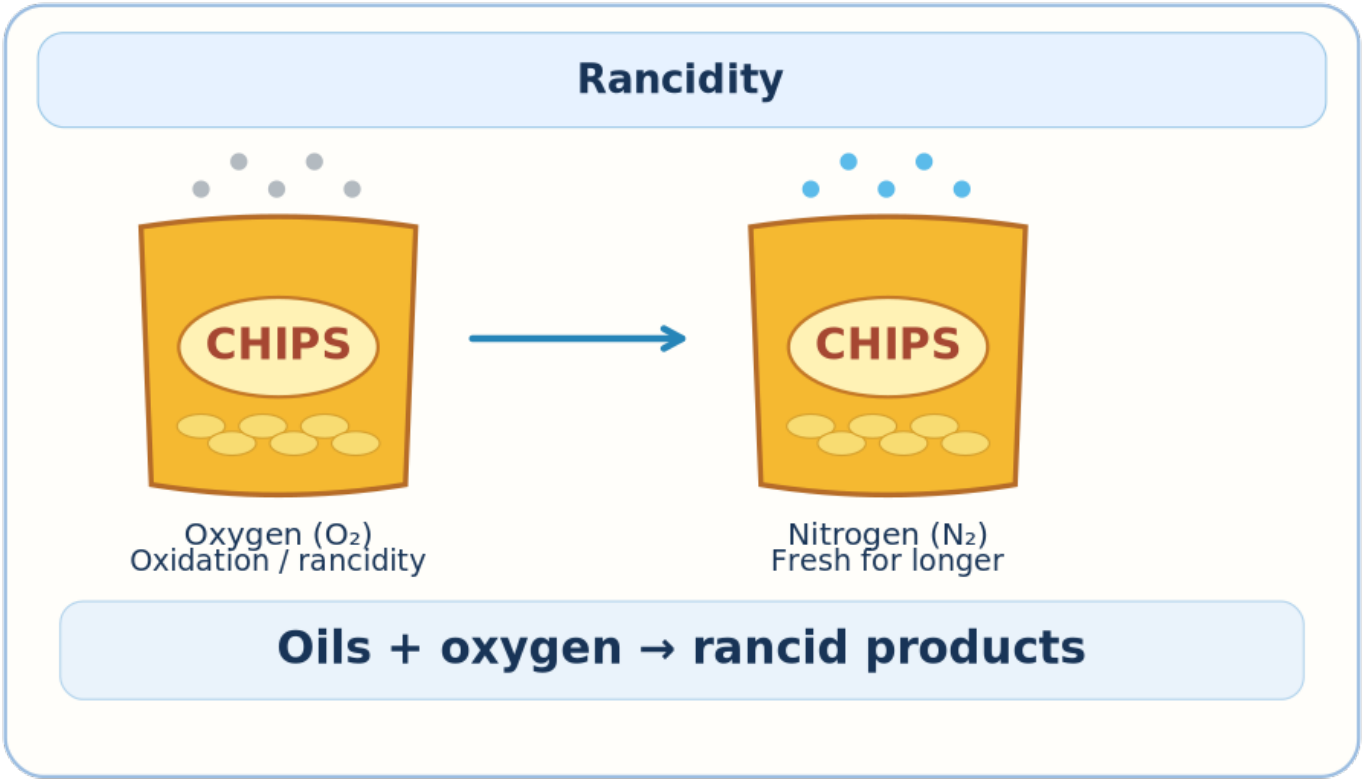
Q51. What happens when lead nitrate solution is mixed with potassium iodide solution? Write the balanced equation.



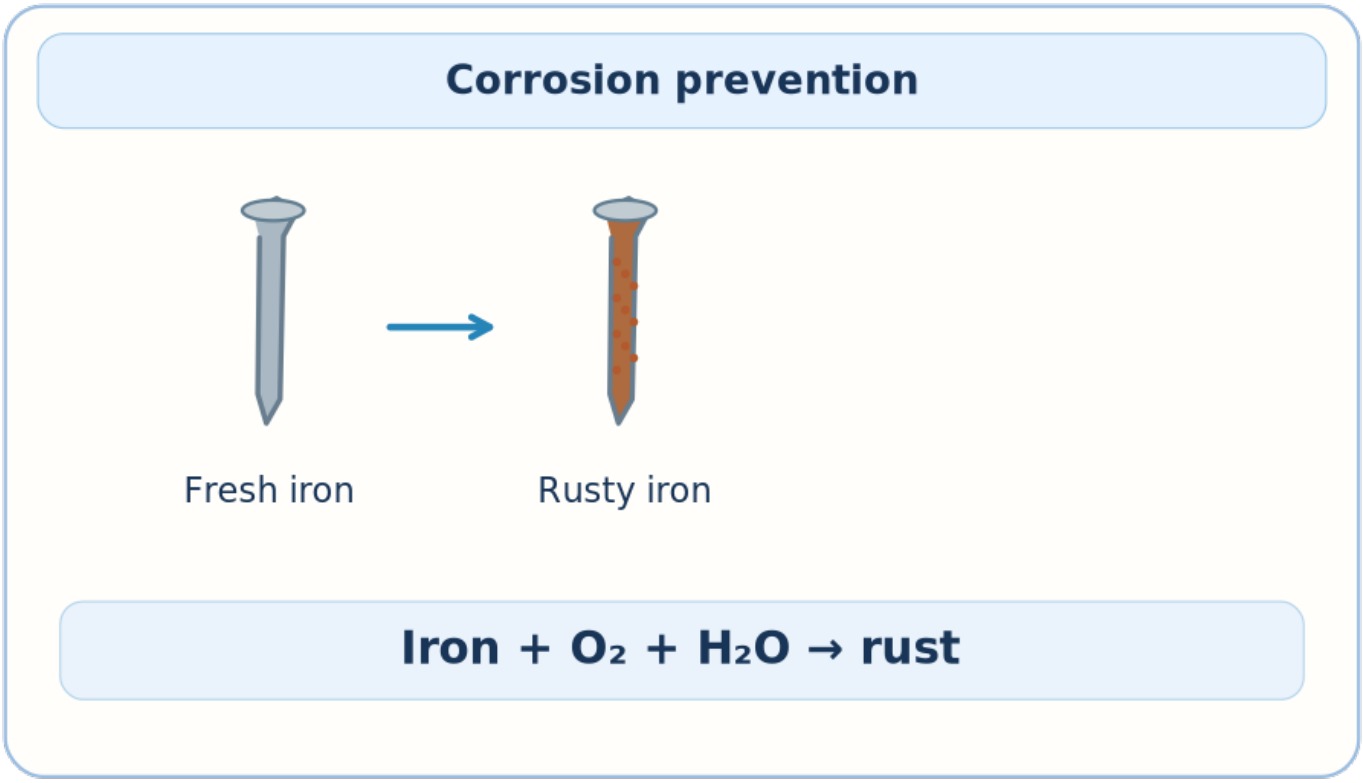
Q52. Why do we store food in a refrigerator?



Q53. What is rancidity and how can it be prevented?



Q54. What causes corrosion of metals? How can it be prevented?



Q55. What is the use of baking soda?

Applications of chemical reactions



Vinegar

CO_2 bubbles ↑

Baking soda reaction

$$2NaHCO_3 \xrightarrow{heat} Na_2CO_3 + CO_2 + H_2O$$

Q56. Why is slaking of lime exothermic? Write the equation.

Exothermic reactions



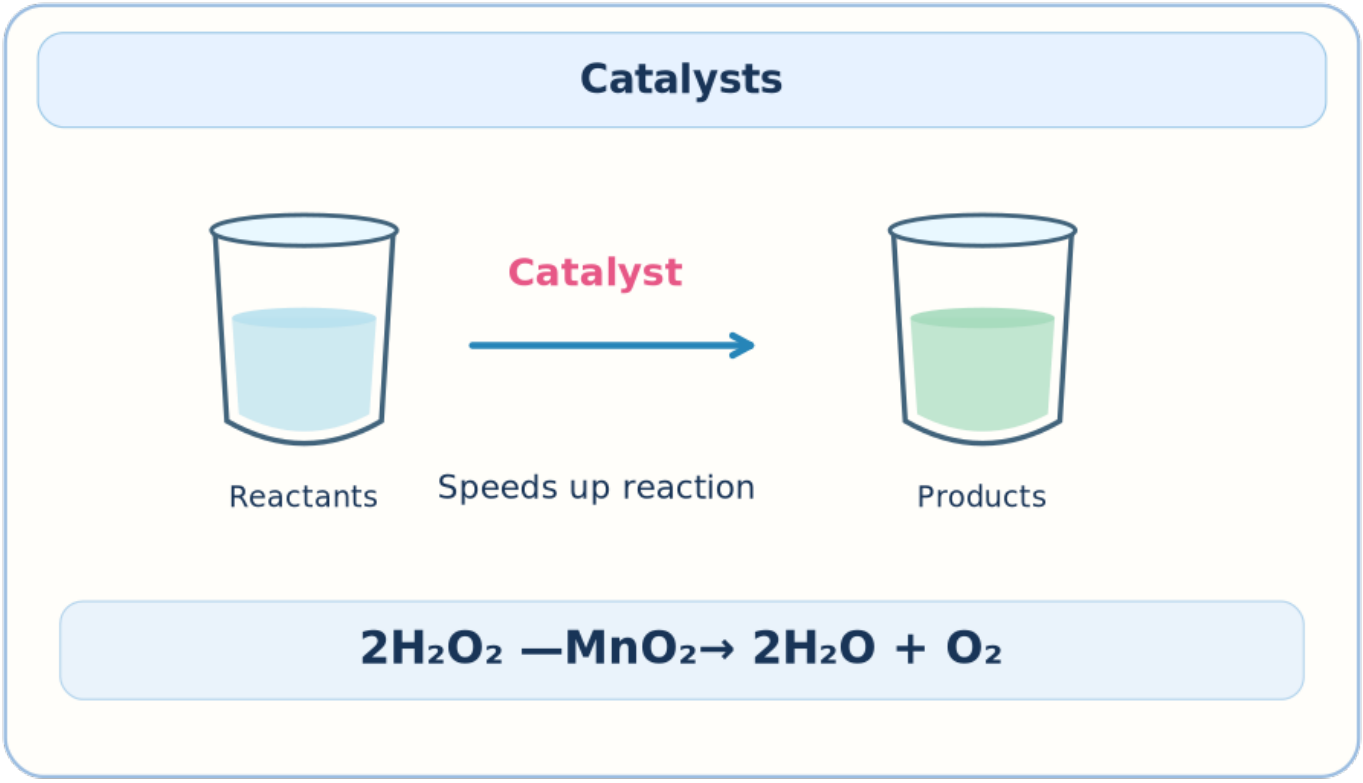
Water

Heat released

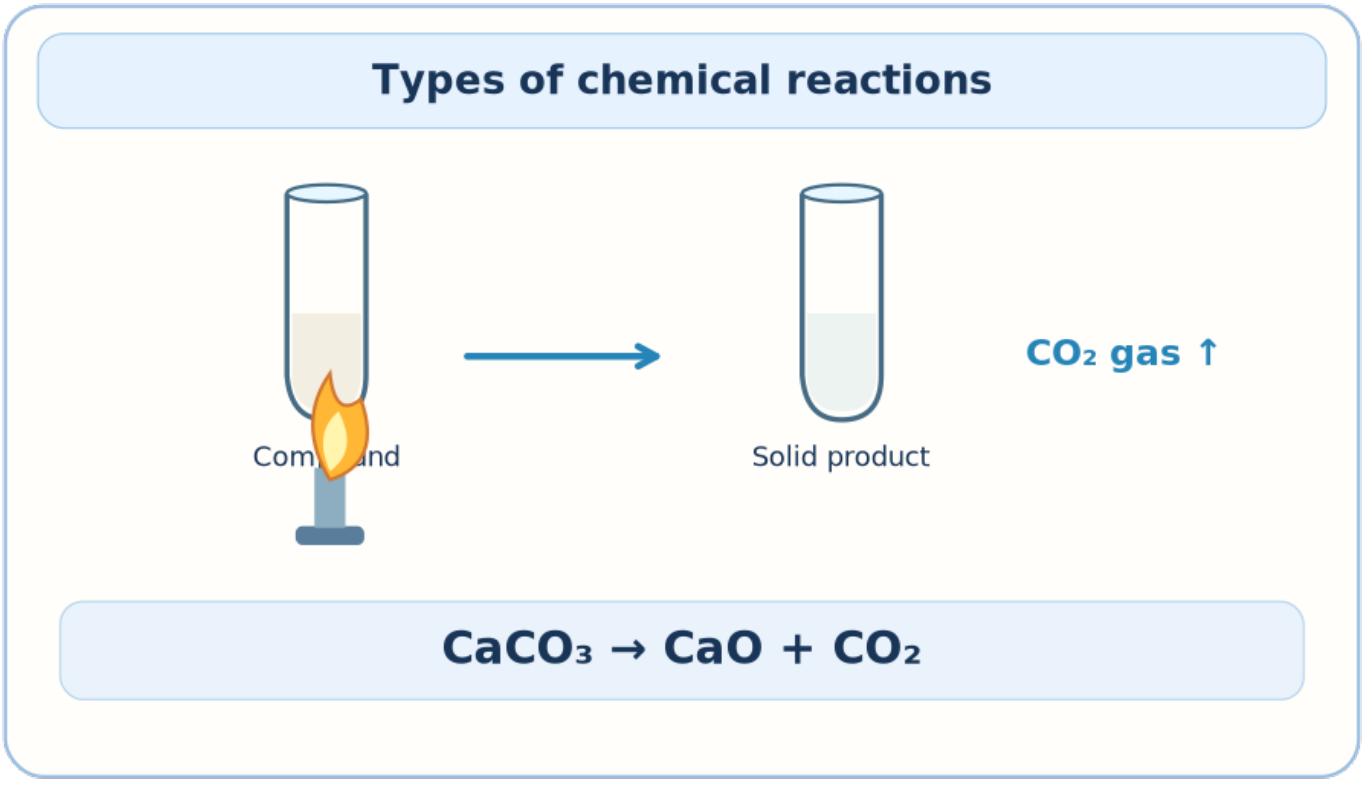
$Ca(OH)_2$

$$CaO + H_2O \rightarrow Ca(OH)_2 + heat$$

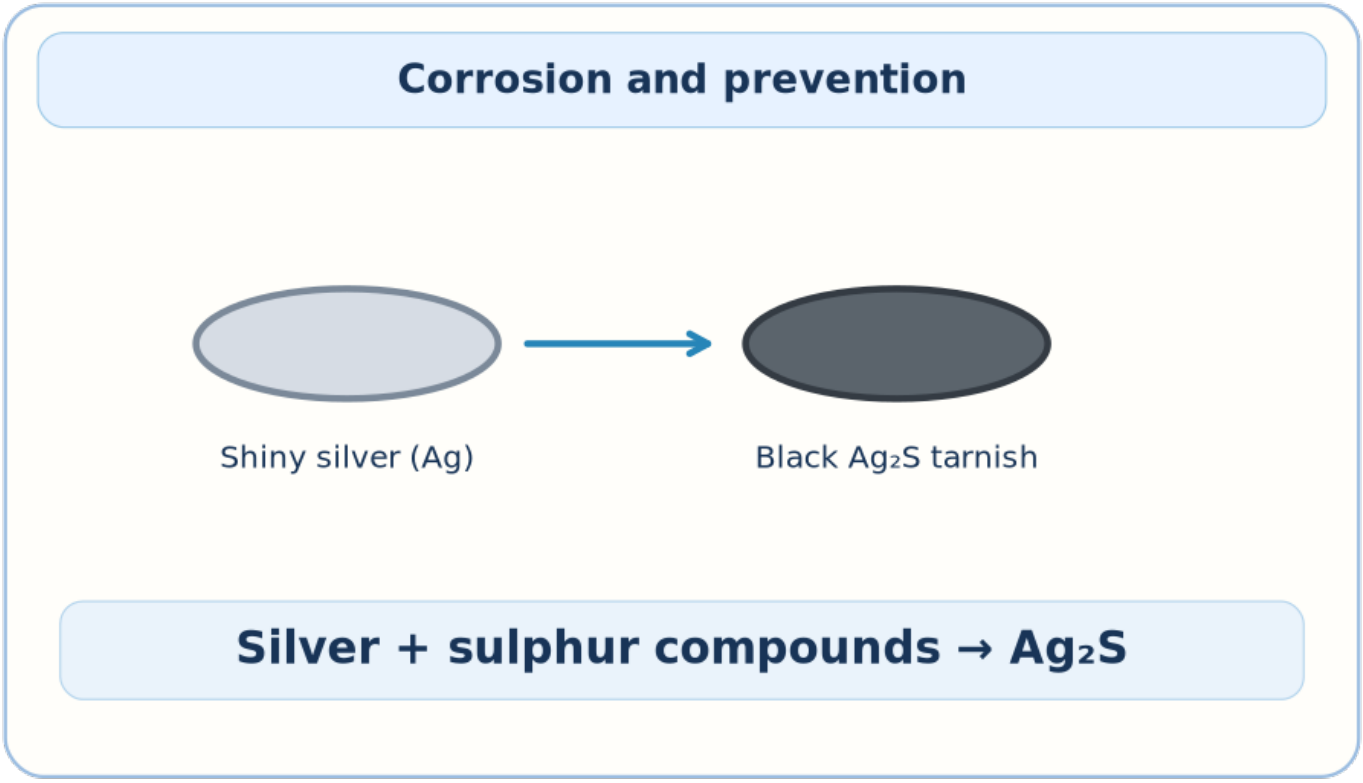
Q57. What is the role of a catalyst? Give an example.



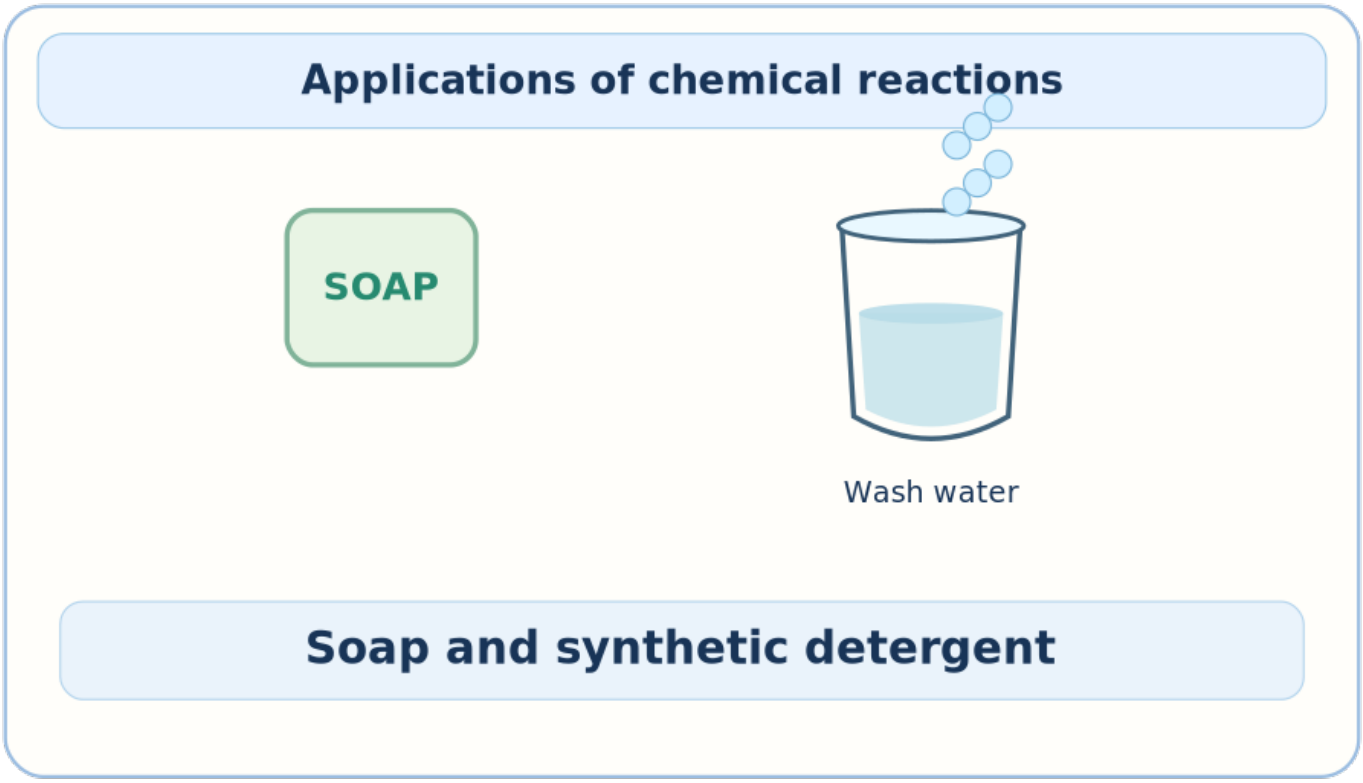
Q58. Differentiate decomposition and double displacement reactions with examples.



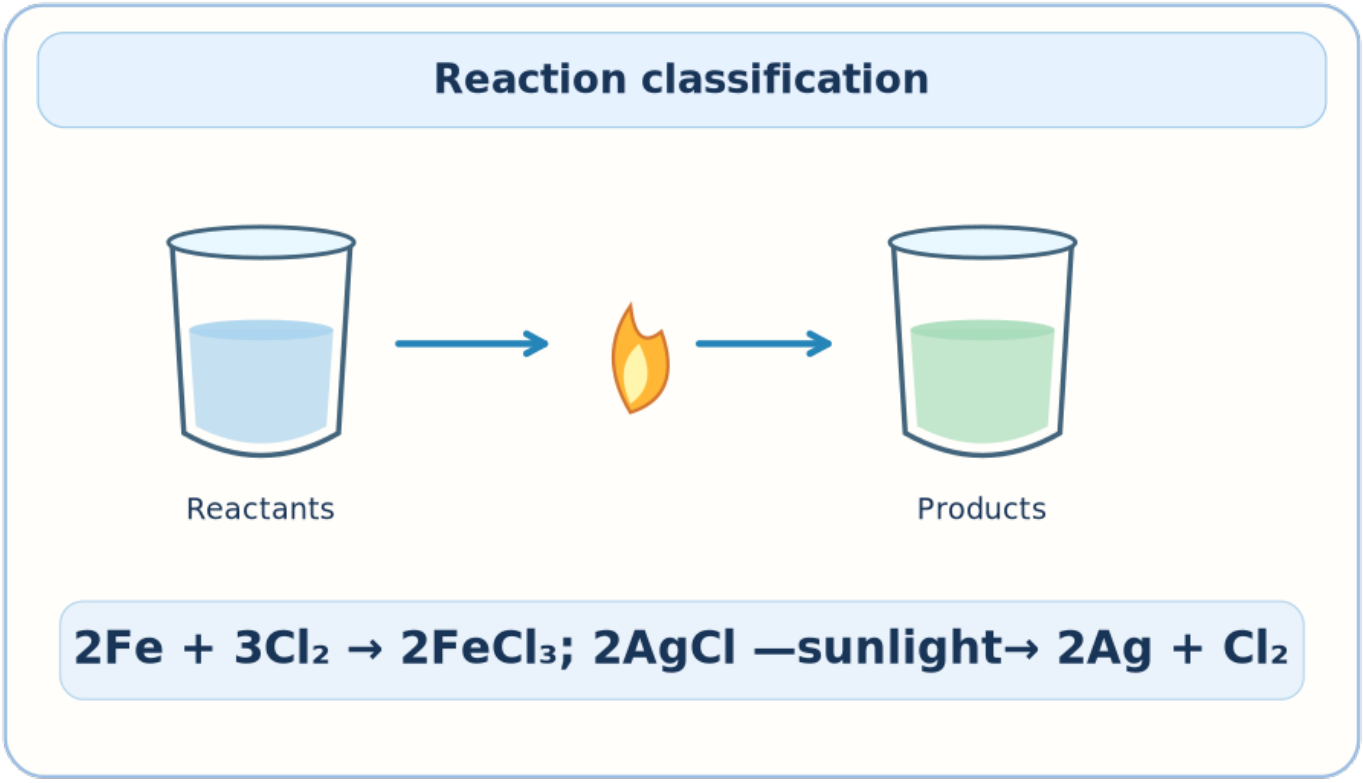
Q59. Why does white coating form on old silver articles? How can it be cleaned?



Q60. Why do we prefer soap over synthetic detergents for washing clothes?



Q61. Identify the type and one benefit of two reactions: iron chloride formation and silver chloride decomposition.



Q62. Write balanced equations for quicklime with water and aluminium with oxygen.

